Dolbeault, Bott–Chern, and deformation invariants of the supplied complex six-sphere families

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Scope and conventions. We give proofs of the statements labelled Proved below and retain explicit unresolved problems under the label Open. The report records completion item by item; in particular a numerical check does not establish a theorem. All coherent sheaves and maps between them are holomorphic, and cohomology has coefficients in $\mathbb{C}$ unless $\mathbb{Z}$ or $\mathbb{Q}$ is written. We use the complex orientation.

Put $B = \mathbb{P}^1_t$, $p_1 = 0$, $p_2 = 1$, $p_0 = \infty$, and $\Sigma = \{p_0, p_1, p_2\}$. For the original family, $f^*p_1 = 3S_1$, $f^*p_2 = 4S_2$, and $W = f^{-1}(p_0)$ is reduced. Its normalization is $\eta : \mathrm{dP}_6 \to W$, with three double curves and two triple points. We distinguish m_jS_j from S_j throughout. The marked lattice is $\Lambda = \mathbb{Z}\langle\hat\gamma, \hat u, \hat w, \hat\delta\rangle$. The standard twists are $v_1 = (1, 2, -4, 0)$, $v_2 = (-1, -3, 3, 0)$, and $(l_0, l_1, l_2) = (0, 1, -1)$, so $p = 12l_0 - 4l_1 - 3l_2 = -1$.

We retain $\Pi = \left(\begin{smallmatrix} 6\mu & \tau & 1 & 0 \\ \beta & \mu & 0 & 1 \end{smallmatrix}\right)$, $\beta_c = \beta^{\mathrm{part}} + c$, and $D = \operatorname{Im}\beta - 6(\operatorname{Im}\mu)^2/\operatorname{Im}\tau$. The admissible range is $\operatorname{Im} c < -M$, where $M = \max D_{\mathrm{part}}$, and $w = e^{-\pi i c}$. Uniformizing coordinates at finite points are stated explicitly when differentiated.

Geometric datum	Period notation	Engel notation
Order-three point	p_1, A_1	0, T_0
Order-four point	p_2, A_2	1, T_1
Cusp	p_0, M_0	∞, T_∞
Elliptic line bundle	L_μ	$M_S = \mathcal{O}_S(P - O)$
Contracting generator	inverse first-period action	lift of $-6P$

The period meridians are clockwise; Engel's are counterclockwise. For column vectors, one integral lattice dictionary to Engel's B_0 basis is

$$Q = \begin{pmatrix} 2 & -1 & 0 & -2 \\ 2 & -1 & 0 & -1 \\ 0 & 2 & 1 & 3 \\ 1 & 0 & 0 & 0 \end{pmatrix}, \quad T_0Q = QA_1^{-1}, \quad T_1Q = QA_2^{-1}, \quad Q\hat\delta = (-2, -1, 3, 0)^{\mathsf{t}}.$$

The cusp path convention gives $QM_0Q^{-1} = T_0^{-1}T_\infty^{-1}T_0$. Inverting the affine deck generators gives $a_0 = -Qv_1/3$, $a_1 = -Qv_2/4$; changing collar sections retains its integral translation. For a reference lift compatible with the period trivialization, $u = e^{-2\pi i c}$; an independently scaled reference has $u = \lambda_0 e^{-2\pi i c}$, $\lambda_0 \in \mathbb{C}^*$. We do not assign a numerical value to that unspecified scale. The exact matrix identities are checked in `code/b_dictionary.py`. No priority is claimed for this AI-produced work.

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1 The cotangent direct images and the decisive period scalar

1.1 A.1. Local logarithmic bundles and a duality calculation

We use the marked threefold of the question, with its arbitrary admissible constant c. Coefficients of coherent cohomology are complex. In this section $\omega_f = K_X \otimes f^*K_B^{-1}$, $\mathcal{L} = \Omega^1_B(\log \Sigma) = \mathcal{O}_B(1)$, $\mathcal{A} = \Omega^1_X(\log D)$, and $\mathcal{E} = \mathcal{A}/f^*\mathcal{L}$. We write γ, u also for their images in $R^1 f_*\mathcal{O}_X$; this convention concerns cohomology classes and does not identify them with differential-form representatives.

Statement A1.0 [Proved]. The structure-sheaf input in the precise form used below.

The direct images $R^q f_*\mathcal{O}_X$, for $q = 0, 1, 2$, are $\mathcal{O}$, $G = \mathcal{O}\gamma \oplus \mathcal{O}(-1)$, and $\mathcal{O}(-1)$. The inclusion $\mathcal{O}\gamma \subset G$ is a subbundle. At the cusp, (γ, u) is a holomorphic frame, and cup product identifies $\bigwedge^2 G$ with $R^2 f_*\mathcal{O}_X$. In the cyclic covering coordinate s, the action on G has characters $(1, \zeta^{-a})$, where $(m, a) = (3, 2)$ or $(4, 1)$ and $\zeta = e^{-2\pi i/m}$. These assertions concern full scheme fibres, including $3S_1$ and $4S_2$.

Proof. We derive the assertion from the local charts and the discriminant; the same direct images are stated in Proposition 9.13 of Claude and Alpöge (2026), but that proposition is not an input to this proof. Flatness holds because a nonzero local parameter on the smooth base is a nonzero divisor in the local ring of the smooth total space. The normalization complex of the cusp has rational surface and curve strata and the cellular cochain complex of a torus as its constant part. It gives $h^q(\mathcal{O}_W) = (1, 2, 1)$; the two degree-one classes γ, u have nonzero cup product. To compute the determinant line without assuming a canonical-bundle formula, we first retain the two finite stabilizers. Let $\mathcal{B}$ be the root curve of orders three and four over B, with root lines R_1, R_2, and put $R = R_1 R_2^{-1}$. The presentation $3R_1 = 4R_2 = H$ gives $\mathrm{Pic}(\mathcal{B}) = \mathbb{Z}R$, $R_1 = R^4$, $R_2 = R^3$ and $\mathcal{O}_{\mathcal{B}}(p_0) = R^{12}$; one may also verify this by removing a line bundle's two stabilizer characters and descending it to $\mathbb{P}^1$.

Write $\bar{f} : X \to \mathcal{B}$. Translation-invariant vertical fields on the torus charts, extended as logarithmic fields on the cusp charts, have transition matrices depending only on the base. They give a rank-two bundle V on $\mathcal{B}$ whose pullback is the relative logarithmic tangent bundle. Its fixed field ξ is a nowhere-zero subline, including in the logarithmic frame at the cusp. The quotient line $Q = V/\mathcal{O}\xi$ has automorphy factors $-1/\tau$ and $1/\tau$ for the two finite generators, read directly from the linear part of the given affine deck transformations. Thus $\Delta(\tau)^{-1}$ is a meromorphic section of Q^{12}. The discriminant product has no zero in the upper half-plane and has a simple zero in the cusp coordinate, since $\tau = (\log t_c)/(2\pi i) + h(t_c)$ with h holomorphic. Its divisor therefore proves $Q^{12} = \mathcal{O}_{\mathcal{B}}(-p_0) = R^{-12}$. The displayed torsion-free Picard group gives $\det V = Q = R^{-1}$.

The exact logarithmic tangent sequence has quotient $\bar{f}^*T_{\mathcal{B}}(-p_0)$. Its determinant, together with $W = \bar{f}^*p_0$, gives $\omega_{X/\mathcal{B}} = \bar{f}^*(\det V)^{-1}$. The geometric fibres on the root charts have cohomology dimensions $(1, 2, 1)$: they are smooth tori at the two finite points, and the normalization calculation above gives the same result at the cusp. Analytic base change makes the three root direct images locally free. Connectedness gives $\bar{f}_*\mathcal{O}_X = \mathcal{O}_{\mathcal{B}}$, and fibrewise Gorenstein duality with its trace pairing now yields $R^2\bar{f}_*\mathcal{O}_X = \det V = R^{-1}$. The invariant local exponents of R^{-1} are two at the order-three point and one at the order-four point. Removing these characters leaves $R^{-1}R_1^{-2}R_2^{-1} = R^{-12}$, so its coarse direct image is $\mathcal{O}_B(-1)$. This supplies the degree used below independently of the asserted direct-image or canonical-bundle formulas in Claude and Alpöge (2026).

The same base-change calculation identifies γ, u with a frame through the cusp. At a multiple fibre finite pushforward and the Reynolds projector identify cohomology with cyclic invariants. If b_0 and b_1 have characters $1, \zeta^{-a}$, their invariant frames are $b_0, s^a b_1$ and $s^a(b_0 \wedge b_1)$.

Their products have exactly these exponents, so the determinant comparison is an isomorphism, including after reduction modulo s^m. Thus $\deg G = -1$. The invariant flat class γ does not enter the holomorphic two-plane spanned by $\sigma_1 = 6\mu\gamma + \tau u + w$ and $\sigma_2 = \beta\gamma + \mu u + \delta$: comparison of the w, δ coefficients proves this even on the smooth cyclic covers at their centres. It therefore gives the stated subbundle everywhere. The quotient is $\mathcal{O}(-1)$ and the extension splits because $H^1(\mathcal{O}(1)) = 0$. This also specifies which subline in the otherwise noncanonical splitting is used below. Coherent Leray now gives $(h^q(\mathcal{O}_X))_{q=0}^3 = (1, 1, 0, 0)$, since $H^0(\mathcal{O}(-1)) = H^1(\mathcal{O}(-1)) = H^1(\mathcal{O}) = 0$. □

Statement A1.1 [Proved]. Local freeness and the determinant line.

Both $\mathcal{A}$ and $\mathcal{E}$ are locally free, of ranks three and two. The vertical vector $\xi = \partial/\partial\zeta_2$ is nowhere zero as a section of $\mathcal{E}^\vee$. There is an exact sequence of holomorphic vector bundles

$$0 \longrightarrow N \longrightarrow \mathcal{E} \xrightarrow{\langle\cdot,\xi\rangle} \mathcal{O}_X \longrightarrow 0, \qquad N = \det \mathcal{E} = \omega_f(-2S_1 - 3S_2). \tag{1}$$

Proof. At a smooth point of a reduced cusp branch the assertion is the usual coordinate calculation. At a double point write $t_c = xy$ with third coordinate z; then $\mathcal{A} = R\,d\log x \oplus R\,d\log y \oplus R\,dz$ and its base subline is $R(d\log x + d\log y)$. At either triple point write $t_c = xyz$; then $\mathcal{A}$ has the three logarithmic coordinate basis vectors and its base subline is their sum. In each case that subline is a direct summand. These calculations apply on every one of the three double curves, including their endpoints; no simplicity of the global divisor is assumed.

At S_j we use the actual free cyclic quotient $q : J_j' \to N_j$, which is finite and etale on total spaces, and $t_j = s^{m_j}$ on the base. On J_j' the logarithmic conormal is $m_j\,ds/s$, and its coefficient is a unit over $\mathbb{C}$. It is again a direct summand of the logarithmic cotangent bundle. The comparison $q^*\mathcal{A} = \Omega^1_{J_j'}(\log(s = 0))$ identifies the quotient bundles and their exact sequence; faithful flat descent proves the claim downstairs. This is not a replacement of $m_j S_j$ by S_j.

On the marked torus family ξ is the constant second vertical vector. The linear monodromies fix it, and translations preserve it. On a cusp toric chart it is a nonzero constant vector in the relative logarithmic tangent lattice of rank two. A toric coordinate change $x_i \mapsto c_i(t_c)t_c^{b_i}x_i$ changes $d\log x_i$ by a multiple of $d\log t_c$, so this constant relative vector descends and is nowhere zero in that bundle. It can vanish as an ordinary tangent vector at a toric stratum; this does not imply vanishing in the logarithmic bundle. Hence evaluation is surjective and its kernel is the determinant line. Taking determinants of $0 \to f^*\mathcal{L} \to \mathcal{A} \to \mathcal{E} \to 0$ gives

$$\det \mathcal{E} = K_X(D) \otimes f^*\mathcal{L}^{-1} = \omega_f(D - f^*\Sigma) = \omega_f(-2S_1 - 3S_2),$$

which proves (1). □

Statement A1.2 [Proved]. The logarithmic direct images by vertical twisting and duality.

One has, as holomorphic vector bundles on B,

q	0	1	2
$R^q f_*\mathcal{E}$	$\mathcal{O} \oplus \mathcal{O}(-1)$	$\mathcal{O}^2 \oplus \mathcal{O}(-1)^2$	$\mathcal{O} \oplus \mathcal{O}(-1)$.

They commute with base change to every full scheme fibre.

Proof. Put $H = 2S_1 + 3S_2$. We first compute the line bundle $\mathcal{O}_X(H)$, rather than a tensor product of two separately descended lattices. On the cover at p_j, its sections allow poles of order at most $m_j - 1$ in s. A fixed character has invariant exponent zero and gains no pole,

whereas the variable character ζ^{-a} has least allowed invariant exponent $a-m$. Thus the inclusion $R^1f_*\mathcal{O} \to R^1f_*\mathcal{O}(H)$ is the identity on $\mathcal{O}\gamma$ and changes s^ab_1 to $s^{a-m}b_1 = t_j^{-1}s^ab_1$ on its quotient. It makes exactly one upper modification at each finite point. The same computation applies to R^2. No modification is made at the cusp or a regular fibre. The result is

$$f_*\mathcal{O}(H) = \mathcal{O}, \qquad R^1f_*\mathcal{O}(H) = \mathcal{O} \oplus \mathcal{O}(1), \qquad R^2f_*\mathcal{O}(H) = \mathcal{O}(1). \tag{2}$$

Here the middle extension splits because $H^1(\mathcal{O}(-1)) = 0$. The local invariant modules are free. Tensoring them with $\mathbb{C}\{t_j\}/(t_j)$ gives invariants of $s^{-(m_j-1)}\mathbb{C}\{s\}/(s^m)$, with the indicated surviving character basis vectors. The Reynolds idempotent is linear over the base, so this calculation also proves the comparison with the nonreduced fibre. At the cusp the line bundle is trivial. Cohomology and base change, applied successively from the top degree, proves the claimed local freeness and comparison for (2).

For clarity the relative duality map we use is the cup product followed by the analytic trace, giving a morphism of bundles

$$R^qf_*(\omega_f \otimes \mathcal{O}(-H)) \longrightarrow \big(R^{2-q}f_*\mathcal{O}(H)\big)^\vee.$$

Every fibre is a compact Gorenstein surface: it is the Cartier fibre in a smooth threefold. Serre duality on that scheme makes the map an isomorphism on its residue field, also for m_jS_j. The constant dimensions just computed imply local freeness on both sides, so Nakayama's lemma makes the bundle map an isomorphism. Therefore

$$(f_*N, R^1f_*N, R^2f_*N) = (\mathcal{O}(-1), \mathcal{O} \oplus \mathcal{O}(-1), \mathcal{O}). \tag{3}$$

This is the explicit reduction of analytic relative duality needed here; the scheme fibre is retained throughout.

The connecting maps of (1) are zero on a regular fibre: its relative cotangent is a trivial bundle and evaluation on a constant nonzero vector splits. Source and target direct images in (3) and Statement A1.0 are vector bundles, so these holomorphic connecting maps, zero on a dense open set, are zero everywhere. We obtain extensions in each degree of the listed R^qN by $R^q\mathcal{O}$. Every resulting pair of line-bundle degrees is zero or minus one; their extension groups are $H^1(\mathcal{O}(-1)), H^1(\mathcal{O}), H^1(\mathcal{O}(1))$, all zero. These extensions give exactly the three asserted bundles. Applying the already proved fibre comparisons to the short exact sequences proves base change for $R^q\mathcal{E}$ as well. This argument uses the vertical twist $\mathcal{O}(H)$ and duality; it does not use the workbench's filtration of $f_*\mathcal{E} \otimes G$. □

Statement A1.3 [Proved]. All absolute cotangent direct images.

For every admissible c, the six direct images are

q	$R^qf_*\Omega^1_X$	$R^qf_*\Omega^2_X$
0	$\mathcal{O}(-2)$	$\mathcal{O}(-2) \oplus \mathcal{O}(-1)$
1	$\mathcal{O} \oplus \mathcal{O}(-1)^2$	$\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$
2	$\mathcal{O} \oplus \mathcal{O}(-1)$	$\mathcal{O}$.

(4)

In particular every listed sheaf is torsion-free.

Proof. We use Statements A2.1, A2.2 and A2.4 below, whose proofs do not use (4). They give $\operatorname{coker}\kappa_0 = \mathbb{C}_{p_1} \oplus \mathbb{C}_{p_2}$, surjectivity of κ_1, and rank one on its global sections. Write $\mathcal{K} = \ker\kappa_1$. The logarithmic conormal sequence then gives

$$f_*\mathcal{A} = \mathcal{O}(1), \quad 0 \to T \to R^1f_*\mathcal{A} \to \mathcal{K} \to 0, \quad R^2f_*\mathcal{A} = \mathcal{O} \oplus \mathcal{O}(-1), \quad T = \mathbb{C}_{p_1} \oplus \mathbb{C}_{p_2}.$$

Let $\nu : \widetilde{D} = \mathrm{dP}_6 \sqcup S_1 \sqcup S_2 \to D$ be normalization. The local bases in Statement A1.1 give the exact residue sequence

$$0 \to \Omega^1_X \to \mathcal{A} \to \nu_* \mathcal{O}_{\widetilde{D}} \to 0.$$

Its three residue direct images are $\mathbb{C}_{p_0} \oplus \mathbb{C}_{p_1} \oplus \mathbb{C}_{p_2}$, $\mathbb{C}_{p_1} \oplus \mathbb{C}_{p_2}$, and zero: the normalization of W is rational, and a bielliptic surface has $(h^0, h^1, h^2)(\mathcal{O}) = (1, 1, 0)$. In degree zero the map from $\mathcal{O}(1)$ is local evaluation with the nonzero multipliers $1, 3, 4$. Its kernel is $\mathcal{O}(-2)$.

We check the torsion cancellation, rather than discarding it. At p_j, in the invariant frame $(b_0, s^a b_1)$ of G, residue of $(dt_j/t_j)(xb_0 + ys^a b_1)$ in $H^1(S_j, \mathcal{O})$ is $m_j x[b_0]$. The residue factors through the base logarithmic line on the covering chart. Naturality of the connecting map consequently gives $\operatorname{res}_1 \kappa_0 = 0$. The determinant of κ_0 has a simple zero, so its reduction has rank one, and its image is exactly the one-dimensional kernel $\mathbb{C}s^a b_1$ of this residue. Hence $\operatorname{coker} \kappa_0 \to H^1(S_j, \mathcal{O})$ is an isomorphism. Both torsion summands cancel, giving the canonical identification $R^1 f_* \Omega^1_X = \mathcal{K}$ and the asserted R^2. The rank-one global map $\mathcal{O}^2 \oplus \mathcal{O}(-1)^2 \to \mathcal{O}$ has a nonzero constant component on a trivial summand. Eliminating all other components by bundle automorphisms gives $\mathcal{K} = \mathcal{O} \oplus \mathcal{O}(-1)^2$.

Finally $\Omega^2_X = (\Omega^1_X)^\vee \otimes K_X$. Relative duality, with the locally free direct images now established, gives

$$R^q f_* \Omega^2_X = \left(R^{2-q} f_* \Omega^1_X\right)^\vee \otimes K_B.$$

Substitution of $K_B = \mathcal{O}(-2)$ gives every entry of the second column. No Euler-characteristic identity is used to select a Hodge number. □

1.2 A.2. An exact modular determinant and two numerical routes

The scalar depends on the fourth root chosen in $t - 1 = s^4$. We make that ambiguity explicit. We choose the local branch for which, on the positive $t > 1$ arc leading to the distinguished cusp, $\tau = iy$ with $y > 1$, and

$$\tau = i + \frac{i}{\pi\sqrt{E_4(i)}} s^2 + O(s^4), \qquad \operatorname{Im} \mu_s(0) > 0. \tag{5}$$

The nonvanishing proved below makes the second condition meaningful; it fixes the remaining sign of s. Equivalently it is the lower cusp-regular zero branch of the Weierstrass function in Statement A2.3. Replacing s by $i^j s$ multiplies the reported derivative by i^{-j}. The workbench instead normalizes $ds/dz(z_2) = 1$; its number is obtained by the chain rule and is not numerically the same unspecified coordinate.

Statement A2.1 [Proved]. The complete raw period determinant.

On the marked upper half-plane,

$$6\mu_z^2 - \tau_z \beta_z = \tau_z^2 \frac{E_4(\tau)^2}{E_6(\tau)}. \tag{6}$$

The right side extends holomorphically through the apparent poles. Its exact orders are two at z_1 and zero at z_2; it has no other zero away from their translates. The logarithmic connecting map $\kappa_0 : f_* \mathcal{E} \to \mathcal{L} \otimes G$ has divisor $\operatorname{divisor}(\det \kappa_0) = p_1 + p_2$.

Proof. Consider the holomorphic quadratic differential $\mathscr{Q} = 6(d\mu)^2 - d\tau\, d\beta$. Direct differentiation of the two affine transformation laws in AC26 Definition 3.1 gives

$$g_j^* \mathscr{Q} = \tau^{-2} \mathscr{Q} \quad (j = 1, 2).$$

In particular g_2^2 fixes τ and sends

$$\mu \longmapsto 1 - \tau - \mu, \qquad \beta \longmapsto \beta + 6\tau + 12\mu - 6,$$

which leaves $\mathscr{Q}$ exactly unchanged. The symbolic residuals of all three identities are zero in `a_calculations.py`. The kernel of $\Delta = \mathbb{Z}/3 * \mathbb{Z}/4 \to \mathbb{Z}/3 * \mathbb{Z}/2 = \mathrm{PSL}_2(\mathbb{Z})$ is the normal subgroup generated by g_2^2. The squared multiplier of $\mathscr{Q}$ is the ordinary modular multiplier $(c\tau+d)^{-2}$, so it is one on every kernel element. Two generic points of the marked cover with the same τ have the same t and differ by an element of Δ; its modular image fixes a generic point, hence is the identity. It follows that $k = \mathscr{Q}/(d\tau)^2$ is a single-valued meromorphic function of τ, transforming as a modular form of weight two. This kernel argument is needed: mere equivariance of the separate period functions would not prove descent.

At z_1, write a linearizing parameter r with $g_1 r = \zeta r$, $\zeta = e^{-2\pi i/3}$. If $\mathscr{Q} = K_r dr^2$ has first nonzero coefficient of order n, its transformation law at the fixed point reads $\zeta^{n+2} = \rho^{-2} = \zeta$. Thus $n \equiv 2 \pmod 3$, and K_r vanishes to order at least two. The map τ is unramified there by AC26 Theorem 3.4(i). At z_2, $\tau - i$ has order two and K_s is holomorphic; therefore k has at most a simple pole at i. Away from these points τ is unramified. At the cusp μ and $\beta + \tau$ are holomorphic in q, so $k = 1 + O(q)$.

The modular function kE_6/E_4^2 is therefore holomorphic at ρ, at i, and at the cusp, as well as everywhere else. The usual zero statements for E_4, E_6 (simple at ρ, i, respectively) are the modular facts used in AC26 Section 3.1. This function descends to the compact modular curve $X(1) = \mathbb{P}^1$, so it is constant. Its cusp value is one. This proves (6) and the exact two local orders, without the workbench's degree count for the coarse determinant line.

For completeness we translate to the actual coarse lattices. Choose cover eigenframes e_0, e_1 of $\mathcal{E}$ with characters $1, \zeta^a$, and b_0, b_1 of G with characters $1, \zeta^{-a}$. Their invariant bases are $e_0, s^{m-a}e_1$ and $b_0, s^a b_1$. The logarithmic base frame is $\ell = dt_j/t_j = m\, ds/s$. Changing determinants from these frames to the marked ones gives

$$\det \kappa_0 = \epsilon(t_j) m^{-2} s^{m+2-2a} K_s, \qquad \epsilon(0) \neq 0.$$

This equality follows by taking the two factors s/m from replacing ds by ℓ, the factor s^{m-a} in the source, and the factor s^{-a} in the target. Thus at $(m, a) = (3, 2)$ the total order in s is $1 + 2 = 3$, and at $(4, 1)$ it is $4 + 0 = 4$. Both coarse zeros are simple. At the cusp, the frame change has no elementary divisor and $dZ/d\tau$ tends to $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, with determinant one. Formula (6) excludes any further zero, proving the asserted divisor. □

Statement A2.2 [Proved]. The evaluated scalar and the two local leading terms.

In the coordinate (5),

$$\mu_s(0) = i\, C_4, \qquad C_4 = \frac{\sqrt{2/3}}{\pi E_4(i)^{1/4}} = \frac{4\sqrt{\pi}}{3^{3/4}\Gamma(1/4)^2} > 0. \tag{7}$$

Numerically,

$$C_4 = 0.23660931408078522468770511345044960629430513408582433721922430 72\ldots.$$

For $r^3 = t$, with $r > 0$ on the arc $0 < t < 1$ and τ travelling from ρ to i along the unit circle, set

$$\lambda = \frac{3e^{5\pi i/6}}{2\pi E_6(\rho)^{1/3}}.$$

Then $\tau = \rho + \lambda r + O(r^2)$ and

$$K_r = -\frac{3i\lambda}{2\pi} r^2 + O(r^3).$$

The leading coefficient has real and imaginary parts

$$\begin{aligned} \mathrm{Re}\, K_2 &= 0.08010221108753827140763650309934451243\ldots, \\ \mathrm{Im}\, K_2 &= 0.13874109940222333968402480721280640441\ldots. \end{aligned}$$

Proof. Ramanujan's differential relations give $E_6'(i) = -\pi i E_4(i)^2$ and $E_4'(\rho) = -(2\pi i/3)E_6(\rho)$. Using $j = 1728E_4^3/(E_4^3 - E_6^2)$, we obtain

$$j(\tau) - 1728 = -1728\pi^2 E_4(i)(\tau - i)^2 + O((\tau - i)^3).$$

Since $j = 1728(1 + s^4)$, the chosen branch is precisely (5). Formula (6) now gives

$$K_s(0) = -\frac{4}{\pi^2\sqrt{E_4(i)}} = 6\mu_s(0)^2,$$

because $\tau_s(0) = 0$. This proves nonvanishing and the first formula in (7); the sign is fixed by the stated coordinate convention. Similarly $j(\tau) = -512i\pi^3 E_6(\rho)(\tau - \rho)^3 + cdots$. This gives the stated λ, and substitution into (6) gives the coefficient $-3i\lambda/(2\pi)$.

There is also a geometric nonvanishing proof independent of this modular determinant. Using the section identification proved in A2.3, set $x = s^2X, y = s^3Y, t = 1 + s^4$. The smooth family becomes

$$Y^2 = X^3 - 3(1 + s^4)^3X + 2(1 + s^4)^4s^2, \qquad P = (0, -\sqrt{2}(1 + s^4)^2s)$$

on the chosen branch. At $s = 0$ its equation is $Y^2 = X^3 - 3X$ and the point is $(0, 0)$. The first variation of the curve itself is zero. In the local coordinate Y at that point, $dX/(2Y) = dY/(3X^2 - 3)$, so the Abel integral of P has derivative $\sqrt{2}/3$. The marked period is $-i\Omega_R$, rather than the positive real period Ω_R: the positive imaginary branch $\tau = iy$, $y > 1$, has elliptic scale $F/s = i/\Omega_R$ at zero. Its period has zero first derivative. Division gives $\mu_s(0) = i\sqrt{2}/(3\Omega_R) \neq 0$. Using $g_2(\mathbb{Z} + i\mathbb{Z}) = (4\pi^4/3)E_4(i)$ and $g_2(\Omega_R(\mathbb{Z} + i\mathbb{Z})) = 12$ gives $\Omega_R = \pi E_4(i)^{1/4}/\sqrt{3}$, which identifies this value with (7). This also explains why choosing the positive real period with a positive real s instead would select the reciprocal branch $\operatorname{Im}\tau < 1$.

Here are the numerical methods actually run. The first evaluates $E_4(i) = 1 + 240\sum_{n=1}^{100}\sigma_3(n)e^{-2\pi n}$ at 100 decimal working digits and inserts it in (7). The second integrates

$$\Omega_R = \frac{\sqrt{2}}{3^{1/4}}\int_0^{\pi/2}(1 - \tfrac{1}{2}\sin^2\theta)^{-1/2}\,d\theta, \qquad C_4 = \frac{\sqrt{2}}{3\Omega_R}.$$

This is the real period of $Y^2 = X^3 - 3X$ for $dX/(2Y)$; substituting the three roots $\sqrt{3}, 0, -\sqrt{3}$ in the elementary elliptic-integral change of variable proves the formula. The beta-integral evaluation of the displayed complete elliptic integral gives the gamma expression in (7). The Fourier and quadrature answers agree with absolute discrepancy $1.79 \cdot 10^{-102}$. As a third check, Newton iteration on the Weierstrass Fourier series with $j(\tau) = 1728(1 + s^4)$, using the two roots adjacent to $(1 - \tau)/2$, gives their symmetric derivative. At $s = 10^{-15}$ it is

$$i\,0.23660931408078522468770511345048081721971218\ldots,$$

with discrepancy $3.13 \cdot 10^{-32}$ and maximum residual below $2.9 \cdot 10^{-101}$. The order-three coefficient was independently evaluated using the divisor-sum expansion of E_6 and its Jacobi theta-constant formula, with agreement to the working precision. The code, all output, and precision are supplied. These are numerical corroborations, not interval certificates; nonvanishing and the exact orders were proved analytically above. For example the elementary bound $\sigma_3(n) \leq n^4$ and $e^{-2\pi} < 1/500$ bounds the omitted Fourier tail after N terms by $250(N + 1)^4 500^{-(N+1)}$; roundoff is separately controlled by the working precision and the independent calculation. □

Statement A2.3 [Proved]. Identification of the elliptic logarithm and the verified zero formula.

For the Weierstrass equation

$$y^2 = x^3 - 3t^3(t-1)x + 2t^4(t-1)^2,$$

the section $[\mu]$ is a generator of its rational section group and equals one of $P_* = (0, \sqrt{2}t^2(t-1))$ and $-P_*$. In particular $\wp(\mu, \tau) = 0$ on every regular fibre. Along the branch in (5), with $x_0 = 1 - 1/t$, the exact branch is

$$\mu = \frac{1-\tau}{2} + \frac{i\sqrt{6}}{3\pi} \frac{x_0^{1/4}\, {}_3F_2(1/3, 2/3, 1; 3/4, 5/4; x_0)}{{}_2F_1(1/12, 5/12; 1; 1 - x_0)}. \tag{8}$$

Consequently $\mu - (2-\tau)/3$ has exact order two at the order-three point in $r^3 = t$.

Proof. We include the comparison, since a matching j-map alone would not identify a section. Write F for the elliptic scale, not the threefold projection. It is specified by

$$F^2 = -\frac{3t(t-1)E_4(\tau)}{\pi^2 E_6(\tau)}, \qquad F/t \longrightarrow i\sqrt{3}/\pi$$

at the distinguished cusp. The local orders are two at $r = 0$ and one at $s = 0$; hence the square root exists on the simply connected marked cover. Its transformation laws are $F(g_1 z) = -F(z)/\tau(z)$ and $F(g_2 z) = F(z)/\tau(z)$. These follow first after squaring by modularity, and the signs are fixed by those local orders. Direct substitution in the cubic identifies its regular fibres with $\mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z})$ by

$$x = F^2 \wp(\nu, \tau), \qquad y = \tfrac{1}{2} F^3 \wp'(\nu, \tau).$$

The transformation laws of μ make this a single-valued section P. Its coordinates are meromorphic at the two finite special points and at the cusp, so are rational functions on $\mathbb{P}^1$.

Every rational section Q admits a global holomorphic elliptic logarithm ν on the marked cover. Indeed the exponential map from the trivial complex line bundle to this smooth elliptic family is a covering map with locally uniformly discrete lattice; pullback over a simply connected base lifts the section. The apparent poles in its projective Weierstrass coordinates at a finite point are meromorphic poles of finite order: the coordinates of Q are rational in t, while F is holomorphic with the finite orders just computed. Multiplying the three homogeneous coordinates by a common power of the local uniformizer and cancelling their common zero gives holomorphic coordinates not all zero at the centre. Their limiting point lies on the closed projective Weierstrass family. This is the valuative criterion for properness over the local analytic discrete valuation ring; it is not an assertion that an arbitrary punctured-disc holomorphic map to a compact space has a removable singularity. We can choose this logarithm bounded at the distinguished cusp. To verify this last assertion, put $v = 1/t$, $X = v^2 x$, $Y = v^3 y$. The equation becomes $Y^2 = X^3 - 3(1-v)X + 2(1-v)^2$. A section cannot meet its nodal point $(X, Y) = (1, 0)$ at $v = 0$: substituting $X = 1 + O(v), Y = O(v)$ leaves nonzero coefficient one in the defining equation at order v. On the smooth nodal locus the coordinate $u = \exp(2\pi i\nu)$ is a nonzero finite coordinate, as follows directly from

$$X = 1 + \frac{12u}{(u-1)^2}, \qquad Y = \frac{12\sqrt{3}u(u+1)}{(u-1)^3}.$$

The normally convergent Weierstrass Fourier series extends this coordinate to a nearby smoothing. The inverse function theorem therefore writes the section as a holomorphic unit

$u(v)$, also near $u = 1$, the point at infinity. A logarithm of that unit fixes the bounded lift. Any two such lifts differ by an integer, since an additional nonzero multiple of τ is unbounded.

For this lift, write the integral transformation errors as

$$\nu(g_1 z) = -\nu/\tau + \alpha_1 + \beta_1(\tau - 1)/\tau, \quad \nu(g_2 z) = \nu/\tau + \alpha_2 - \beta_2/\tau.$$

Bounded cusp regularity and $g_1 g_2 = g_0^{-1}$ imply $\alpha_2 = -\beta_1$, $\beta_2 = \alpha_1 + \beta_1$. Putting $n = -\alpha_1 - 2\beta_1$, $a = \alpha_1 + \beta_1$, we find that $\nu - (n\mu + a)$ satisfies the homogeneous law of AC26 Lemma 3.10 and is cusp-regular. We recall a direct verification of the degree used here. The function $F_h = E_4(\tau)^2 E_6(\tau)^{1/2}/\Delta(\tau)$ on the marked cover has orders two and one at z_1, z_2, and has a simple pole at the cusp. The square root exists because its zeros on this cover have even order. Modularity and these local orders fix the two signs: $F_h(g_1 z) = -F_h/\tau$, $F_h(g_2 z) = F_h/\tau$. For any holomorphic solution of the same homogeneous law, the quotient by F_h is an invariant meromorphic function of t. A possible pole at the order-three point would have order at most $2/3$ in t, and one at the order-four point at most $1/4$; its integral order is therefore nonnegative. At the cusp regularity requires this quotient to vanish. Thus the solution sheaf is exactly $\mathcal{O}_B(-p_0)$, with no nonzero global sections. We obtain $\nu = n\mu + a$ and hence $Q = nP$. The two distinct finite values of μ exclude $nP = 0$ for $n \neq 0$ by cusp boundedness; there is no torsion. This proves that P generates the section group.

We identify P_* with a primitive generator by an intersection calculation. The rational elliptic surface and its fibre types $\mathrm{IV}^*, \mathrm{III}, I_1$ are those in Engel Section 2. At $t = 1$, the blowup chart $x = (t-1)X, y = (t-1)Y$ gives $Y^2 = (t-1)X^3 - 3t^3X + 2t^4$; P_* meets its exceptional multiplicity-one component at $(X, Y) = (0, \sqrt{2})$, away from the identity component. At $t = 0$, two successive charts $x = t^2X, y = t^2Y$ give $Y^2 = t^2X^3 - 3t(t-1)X + 2(t-1)^2$; P_* meets the smooth exceptional component $Y = -\sqrt{2}$ at $X = 0$. These are the two nonidentity simple components of the E_6 configuration. The section is disjoint from O in every chart, including infinity.

Here the height calculation can be made directly, without assuming primitivity. Orthogonally project $Q - O$ in $\mathrm{NS}(S) \otimes \mathbb{Q}$ off O, the fibre F_S, and the nonidentity components. The inverse intersection matrices are $-C_{E_6}^{-1}$ and $-1/2$. Their diagonal entries for the two nonidentity simple components are $-4/3$ and $-1/2$. Here we may number the nonidentity components so that

$$C_{E_6} = \begin{pmatrix} 2 & -1 & 0 & 0 & 0 & 0 \\ -1 & 2 & -1 & 0 & -1 & 0 \\ 0 & -1 & 2 & -1 & 0 & 0 \\ 0 & 0 & -1 & 2 & 0 & 0 \\ 0 & -1 & 0 & 0 & 2 & -1 \\ 0 & 0 & 0 & 0 & -1 & 2 \end{pmatrix}, \qquad (C_{E_6}^{-1})_{44} = (C_{E_6}^{-1})_{66} = 4/3.$$

Since $K_S = -F_S$, adjunction gives $Q^2 = O^2 = -1$. Expanding the square of this projection gives

$$h(Q) = 2 + 2(Q \cdot O) - c_0(Q) - c_1(Q), \quad c_0(Q) \in \{0, 4/3\},\ c_1(Q) \in \{0, 1/2\}.$$

The projection is additive: the chord-and-tangent law makes $Q + R - (Q+R)_{\text{section}} - O$ linearly equivalent to a vertical divisor, which the projection kills. Its negative square therefore satisfies $h(nQ) = n^2 h(Q)$. A section meets a single simple component, since its base parameter has order one; consequently the listed local corrections exhaust the possibilities. For $Q \neq O$ we have $Q \cdot O \geq 0$, so $h(Q) \geq 1/6$. For P_* the explicit incidences give $h(P_*) = 1/6$. Writing $P_* = nP$ now gives $1/6 = n^2 h(P) \geq n^2/6$, and hence $n = \pm 1$. Inverting the displayed E_6 Cartan matrix and the chord-and-tangent formulas is included in the exact computation log. This proves $\wp(\mu, \tau) = 0$.

We checked the original theorem on page 399 of Eichler and Zagier (1982), as well as equation (6) of Duke and Imamoğlu (2008). With the normalized Δ, their coefficient is indeed $144\pi i\sqrt{6}$, their constant is $\log(5+2\sqrt{6})/(2\pi i)$, and the integrand is $(\sigma-\tau)\Delta(\sigma)/E_6(\sigma)^{3/2}$ on the vertical path with its continued branch. Thus the formula in Lead 3 is correct with its indicated sign and period ambiguity. Theorem 1 of Duke–Imamoglu gives (8): we take one minus their z_0, rather than $z_0-\tau$, because the latter is unbounded at the distinguished cusp. The chosen branch approaches $(1-i)/2$ at $\tau=i$ and has lower imaginary cusp value. These conditions identify the original bounded lift. In particular its linear coefficient is $i\sqrt{6}/(3\pi\,{}_2F_1(1/12,5/12;1;1))$, agreeing with (7).

Finally exact elliptic addition gives

$$x(2P_*)=9t^2/8,\quad y(2P_*)=\frac{t^2(32-5t)}{16\sqrt{2}},\quad x(3P_*)=\frac{16}{81}(t-1)(5t-32).$$

The last expression has nonzero limit $512/81$ at $t=0$. If $\nu=\mu-(2-\tau)/3$, then 3ν is the local logarithm of $3P$ modulo its exact period. Since F has order two and $\wp(3\nu)=(3\nu)^{-2}+O(\nu^2)$, the equality $x(3P)=F^2\wp(3\nu)$ forces ν to have exact order two. It also gives its leading coefficient $\nu_2=\pm3F_2/(16\sqrt{2})$, where $F_2=\lim_{r\to0}F/r^2$; the sign is the explicitly stated elliptic-logarithm branch. □

Statement A2.4 [Proved]. The global rank and Lead 2.

The map $\kappa_1:R^1f_*\mathcal{E}\to\mathcal{L}\otimes R^2f_*\mathcal{O}=\mathcal{O}$ is surjective, and $H^0(\kappa_1):\mathbb{C}^2\to\mathbb{C}$ has rank one. Let $h=a\,d\zeta_1+d\zeta_2$ be the unique global section of $f_*\mathcal{E}$ with $h(\xi)=1$. In a genuine invariant target frame normalized as below,

$$\kappa_1(h\gamma)(p_2)=-\tfrac{1}{4}\mu_s(0)=-\tfrac{i}{4}C_4.$$

The rank-one condition is equivalent to $a\tau_z+\mu_z\not\equiv0$. Under the same direct-image and sheaf-surjectivity hypotheses a rank-zero global map would instead give $h^{1,2}=2$, $h^{1,1}=3$; that alternative does not occur here.

Proof. Let $d_{ij}(s)$ be the cover Kodaira–Spencer coefficients in the eigenframes of Statement A2.1 after removing ds. The invariant coarse matrix and its cup-product row have the forms

$$[\kappa_0]=\begin{pmatrix}A & t_jB\\ C & D_0\end{pmatrix},\qquad [\kappa_1]=(-C,A,-D_0,B),$$

in the ordered basis $e_0b_0,s^ae_0b_1,s^{m-a}e_1b_0,e_1b_1$. Specifically $A=sd_{00}/m$, $B=s^{1-a}d_{01}/m$, $C=s^{1-a}d_{10}/m$, $D_0=s^{m+1-2a}d_{11}/m$. These are holomorphic invariant functions; the row follows by wedging the connecting class with b_0,b_1. If the row had no unit at a finite point, every entry would vanish modulo t_j and $\det\kappa_0$ would vanish to order at least two, contradicting A2.1. Away from the finite points, including the cusp, κ_0 is invertible, which gives a unit in the cup-product row. Thus κ_1 is surjective.

The lift h exists and is unique because $f_*\mathcal{E}$ is an extension of $\mathcal{O}$ by $\mathcal{O}(-1)$. Its invariance under g_2 gives $a(g_2z)=\tau a+\mu$, hence $a(z_2)=1/2$. In the quotient frame (γ,u) of G the connecting map is

$$\kappa_0(h)=\frac{s}{4}((6a\mu_s+\beta_s)\gamma+(a\tau_s+\mu_s)u)\ell.$$

Choose the variable eigenframe b_1 by Reynolds averaging so that $\gamma\wedge b_1=v(s)\gamma\wedge u$, $v(0)=1$. Then $s\,v(s)(\gamma\wedge u)\ell$ is the actual invariant target frame; the uncorrected expression with $v=1$ need not be invariant away from zero. Cup product with γ gives at the central point $-(a\tau_s+\mu_s)/4=-\mu_s/4$, proving rank one.

To check the stated equivalence, there is a second global section

$$q_E = \sigma_1 \otimes u + 6\sigma_2 \otimes \gamma \quad \text{of } R^1 f_* \mathcal{E}.$$

It is the negative of the relative $(1,1)$ image of the invariant flat class $\eta = u \wedge w + 6\gamma \wedge \delta$. It extends at the cusp because the relative logarithmic forms and γ, u extend there. At a finite point it is an invariant holomorphic tensor on the smooth cover, hence belongs to the four explicit invariant generators above. Its cup-product row is zero, since the two terms are $6\mu_z$ and $-6\mu_z$. It is independent of $h\gamma$ on every regular fibre (its $\sigma_1 \otimes u$ coefficient is nonzero), so these two sections form a basis of the two-dimensional global section space. Consequently the global map has rank one precisely when its value on $h\gamma$, equivalently $a\tau_z + \mu_z$, is not identically zero. If its rank were zero, the long exact cohomology sequence of $0 \to \mathcal{K} \to \mathcal{O}^2 \oplus \mathcal{O}(-1)^2 \to \mathcal{O} \to 0$ would give $h^0(\mathcal{K}) = 2$, $h^1(\mathcal{K}) = 1$. The residue calculation in A1.3 and Leray would then give exactly the two counterfactual Hodge numbers stated in the claim. □

1.3 A.3. The Hodge table and its Leray generators

Statement A3.1 [Proved]. The complete Hodge table for every admissible parameter.

For every $\operatorname{Im} c < -M$, the Hodge numbers are

$q\backslash p$	0	1	2	3
3	0	0	0	1
2	0	1	2	1
1	1	2	1	0
0	1	0	0	0.

Proof. Since B is a curve, coherent Leray has no differential and gives for $F = \Omega_X^p$ the exact sequences $0 \to H^1(B, R^{q-1} f_* F) \to H^q(X, F) \to H^0(B, R^q f_* F) \to 0$. Substitution of (4), with $h^0(\mathcal{O}) = h^1(\mathcal{O}(-2)) = 1$ and all cohomology of $\mathcal{O}(-1)$ zero, gives columns $p = 1, 2$ directly. Statement A1.0 gives column zero, and Serre duality gives column three. Thus the numbers are outputs of the sheaf computation. Replacing β by $\beta + c$ leaves every derivative, finite eigencharacter, invariant lattice and logarithmic residue used above unchanged. The toric gluing units change, but their relative logarithmic derivatives still differ only by base logarithmic forms. Hence the same computations apply at each admissible c, including every point arbitrarily close to the strict boundary; no manifold at the excluded boundary is asserted. □

Statement A3.2 [Proved]. Specified associated-graded generators and a global Dolbeault lift.

The Leray associated-graded spaces for the four middle groups are

	$E_2^{1,q-1}$	$E_2^{0,q}$
$H^{1,1}$	$H^1(\mathcal{O}(-2)) = \mathbb{C}\alpha$	$\mathbb{C}\eta_{\mathrm{rel}}$
$H^{1,2}$	0	$\mathbb{C}\upsilon$
$H^{2,1}$	$\mathbb{C}\upsilon^\vee$	0
$H^{2,2}$	$\mathbb{C}\eta_{\mathrm{rel}}^\vee$	$\mathbb{C}\alpha^\vee$.

Here α is the Cech class of dt/t on the two standard affine base charts, $\eta_{\mathrm{rel}} = -q_E$ is the invariant relative class from A2.4, and υ is the image of the trace-dual of $1 \in f_*\mathcal{O}(H)$ under $R^2 f_* N \to R^2 f_* \mathcal{E}$ and the isomorphism $R^2 f_* \Omega_X^1 = R^2 f_* \mathcal{E}$. The superscript $\vee$ denotes the

bases dual under the relative trace and the base Serre pairing. There exists a Dolbeault class $Q \in H^1(X, \Omega^1_X)$ whose restriction to every regular fibre is $\eta = u \wedge w + 6\gamma \wedge \delta$.

Proof. The isomorphism $f_*\Omega^1_X = \Omega^1_B$ in the residue sequence is the actual pullback map, so α gives its stated Leray class; it can be represented by a nonzero normalization of $f^*\omega_{\mathrm{FS}}$ in Dolbeault cohomology. The section η_{rel} belongs to the canonical kernel identification $R^1 f_*\Omega^1_X = \ker \kappa_1$ and spans its global sections. Leray is surjective onto this space, so any preimage defines Q with the asserted fibre restriction. Its ambiguity is exactly $\mathbb{C}\alpha$; the statement does not claim a canonical global form. The injection $R^2 f_* N = \mathcal{O} \to R^2 f_* \mathcal{E}$ from (1) spans its global sections, since the quotient is $\mathcal{O}(-1)$. This defines υ with no choice of a negative degree summand. Duality used in A1.3 gives the remaining entries and their specified dual pairings. Choosing a lift of η_{rel} and of $\alpha^\vee$ gives bases of the whole middle cohomology groups. The associated-graded descriptions are canonical with the named choices; the splitting of Leray vector spaces is not asserted canonical. □

1.4 A.4. Ordinary relative differentials and the correction at the cusp

Statement A4.1 [Proved]. All local modules needed to distinguish degrees one and two.

Let $R = \mathbb{C}\{x, y, z\}$. For $t = xy$,

$$\Omega^1_{R/\mathbb{C}\{t\}} = R^3/R(y, x, 0) \simeq (x, y) \oplus R,$$

and

$$\Omega^2_{R/\mathbb{C}\{t\}} \simeq (x, y) \oplus R/(x, y), \qquad \Omega^3_{R/\mathbb{C}\{t\}} \simeq R/(x, y).$$

For $t = xyz$, $\Omega^1 = R^3/R(yz, xz, xy)$ is torsion-free of rank two and is not free at the triple point. If $I = (xy, xz, yz)$, there is an exact sequence

$$0 \to T \to \Omega^2 \xrightarrow{\wedge dt} I \to 0, \qquad T = \frac{Ru \oplus Rv}{(xu,\ y(u - v),\ zv)}, \quad IT = 0,$$

and $\Omega^3 = R/I$. Finally, for $t = s^m$ in $R = \mathbb{C}\{s, y_1, y_2\}$,

$$\begin{aligned} \Omega^1 &= R^2 \oplus R/(s^{m-1})\, ds, \\ \Omega^2 &= R\, dy_1 \wedge dy_2 \oplus \left(R/(s^{m-1})\right)^2, \\ \Omega^3 &= R/(s^{m-1})\, ds \wedge dy_1 \wedge dy_2. \end{aligned}$$

Proof. The degree-one presentation is the quotient by dt; higher degrees are its exterior powers. At a double point the transverse quotient $R^2/R(y, x)$ maps isomorphically to (x, y) by $(a, b) \mapsto xa - yb$ after an immaterial change of sign of one basis vector. More explicitly one may use $(a, b) \mapsto xa - yb$ with relation (y, x); its kernel is precisely that relation, since x, y are relatively prime. In degree two, with basis $A = dx \wedge dy, B = dx \wedge dz, C = dy \wedge dz$, the relations are $xA = yA = 0$, $yB + xC = 0$. This gives the displayed torsion summand and the ideal (x, y); wedging once more gives $R/(x, y)$.

At the triple point the map from ordinary to logarithmic relative one-forms sends (a, b, c) to $(xa - zc, yb - zc)$. Its kernel is precisely $R(yz, xz, xy)$: the two equations imply $c = xyh$, $a = yzh$, $b = xzh$ by relative primality of the coordinates. Thus it injects into R^2, proving torsion-freeness. Its fibre at the maximal ideal has dimension three although its rank is two, so it is not free. The same minimal-generator argument works at the double locus.

In degree two, wedging with dt is the row $(xy, -xz, yz)$ in the basis (A, B, C). Its syzygies are generated freely by $u = (z, y, 0)$ and $v = (0, y, x)$. To check generation, an equation $xyA - xzB + yzC = 0$ implies $x \mid C$, say $C = xC_1$; then $yA = z(B - yC_1)$ implies $A = zA_1$, $B = y(A_1 + C_1)$. Thus the syzygy is $A_1u + C_1v$. The three relations from $dt \wedge (dx, dy, dz)$ are $-xu, y(u - v), zv$, proving the stated presentation of the kernel. The ideal I kills it by those relations. Conversely the quotient I is torsion-free, so this kernel is exactly the torsion of Ω^2. The presentation immediately gives $\Omega^3 = R/I$. All matrix identities have also been checked over $\mathbb{Z}[x, y, z]$ by the supplied exact program. For $t = s^m$ the sole degree-one relation is $ms^{m-1}ds = 0$; m is a unit and taking exterior powers gives the three formulas. □

Statement A4.2 [Refuted]. The claimed degree-one torsion along the cusp.

The assertion in AC26 Remark 9.21, PDF page 81, that $\Omega^1_{X/B}$ has torsion along $W \cup S_1 \cup S_2$ is false. Its torsion is supported precisely on $S_1 \cup S_2$; its degree-one cusp modules are torsion-free but not locally free on the double locus. The plural statement in Remark 1.3, PDF page 5, must be qualified by degree: $\Omega^2_{X/B}$ does have torsion on the cusp double locus. The workbench's degree-one correction is correct; it must not be extended to degree two.

Proof. Both disputed sentences were checked on rendered pages of the original PDF, so this is not a transcription correction. The double and triple charts in A4.1 exhaust the singular strata of W. They prove absence of degree-one torsion at the cusp and exhibit nonzero degree-two torsion there. At a multiple fibre the finite etale comparison $q^*\Omega^1_{N_j/B} = \Omega^1_{J'_j/B}$ is an isomorphism, and its last summand in A4.1 is nonzero for $m = 3, 4$. Faithful flatness descends both its existence and its support. There are no other critical fibres and the ordinary relative cotangent is locally free on their complement. Thus the stated support is exact. The erroneous sentence is at transcription line 4330 (the final paragraph of Remark 9.21); the workbench's correct degree-one replacement occupies lines 980–1006 of `hodge_frolicher.tex`. No numerical Hodge value in that workbench is refuted by this calculation. □

2 B. Other logarithmic transformations and finite covers

2.1 B.1. Comparison under logarithmic transformation

Statement B1.0 [Proved]. The marked geometric dictionary on the first page.

The lattice matrix, meridian orientations, line bundle and parameter comparison stated on the first page identify the two marked smooth families. The ambiguity in u is exactly multiplication by the nonzero constant specifying Engel's reference linearization.

Proof. First quotient the period presentation by its last three columns, using $z = \zeta_1$ and $v = e^{2\pi i\zeta_2}$. The resulting principal $\mathbb{C}^*$-bundle over $\mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z})$ has factors

$$(z, v) \mapsto (z + 1, v), \qquad (z, v) \mapsto (z + \tau, e^{2\pi i\mu}v).$$

For an odd theta function ϑ, the meromorphic function $\vartheta(z - \mu)/\vartheta(z)$ has precisely these factors and divisor $(\mu) - (0)$. Thus this is the complement of the zero section of $L_\mu = \mathcal{O}(P - O)$, with $P = \mu$ on the elliptic curve. The first period column acts by $(z, v) \mapsto (z + 6\mu, e^{2\pi i\beta}v)$; its inverse covers $-6P$, as does Engel's contracting generator. Replacing β^{part} by $\beta^{\mathrm{part}} + c$ multiplies this inverse lift by $e^{-2\pi ic}$. A change of reference lift multiplies its scale by a single constant $\lambda_0 \in \mathbb{C}^*$, proving the asserted parameter comparison and naming its entire ambiguity.

For the integral markings we apply the displayed matrix Q to column lattice vectors. Exact integer multiplication gives $\det Q = -1$, $T_0Q = QA_1^{-1}$ and $T_1Q = QA_2^{-1}$, as recorded with the

matrices in `b_dictionary.py`. These inverses are required because the source meridians have opposite orientations. The product of the meridians gives $QM_0Q^{-1} = T_0^{-1}T_\infty^{-1}T_0$ with the stated cusp path. Finally $A_jv_j = v_j$, so inversion of an affine deck generator with translation v_j/m_j changes that translation to $-v_j/m_j$. Applying Q gives $a_0 = -Qv_1/3$ and $a_1 = -Qv_2/4$. This calculation identifies marked lattices and affine deck actions; it does not discard integral collar translations as if they were equal to zero. □

Statement B1.1 [Proved]. The relative cohomology comparison.

Fix the period family and cusp filling. Let X' be obtained by any admissible translation twists and multiplicities $m_1 = 3k$, $m_2 = 4l$, where k, l are positive integers. Write $D' = W + S_1' + S_2'$ and

$$E' = \Omega^1_{X'}(\log D')/f'^*\Omega^1_B(\log \Sigma).$$

The identity on the common smooth family induces isomorphisms on B

$$R^qf_*\mathcal{O}_X \simeq R^qf'_*\mathcal{O}_{X'}, \qquad R^qf_*E \simeq R^qf'_*E' \quad (0 \le q \le 2).$$

These isomorphisms identify the logarithmic connecting maps κ_q. They also hold when the cusp collar is changed by an admissible translation.

Proof. We construct the comparison at every point. Over $B \setminus \Sigma$ the two families are identified, and a translation acts trivially on $H^q(F, \mathcal{O}_F)$ and $H^q(F, \Omega^1_F)$ for a complex torus F. For the second assertion, translation fixes the invariant holomorphic one-forms, and the first follows either from invariant Dolbeault forms or the exterior algebra $H^\bullet(F, \mathcal{O}_F) = \bigwedge^\bullet H^1(F, \mathcal{O}_F)$. These are statements about the induced maps on cohomology, including their holomorphic variation.

At a point with original order $d = 3$ or 4, let $\widetilde{f} : T \to \Delta_z$ be the smooth proper torus model with coarse coordinate $q = z^d$. The new model is its pullback by $z = s^e$, where $m = de$, followed by the free cyclic quotient of order m. Smooth proper base change identifies the higher direct images upstairs with the pullbacks of those for T. The relative logarithmic cotangent bundle upstairs is precisely Ω^1_{T/Δ_z} pulled back: the base logarithmic direction is killed in the quotient defining E'. The derivative of a translation contributes only a multiple of ds; it consequently acts on this quotient as ordinary translation on relative forms.

Average over the cyclic action and choose holomorphic eigenframes. If a frame for $H^q(\mathcal{O})$ or $H^q(\Omega^1)$ has character ζ_d^{-a}, $0 \le a < d$, its original invariant extension is $z^a\mathcal{O}_{\Delta_q}$. On the order-$m$ cover its character is ζ_m^{-ea}, and its invariant extension is $s^{ea}\mathcal{O}_{\Delta_q} = z^a\mathcal{O}_{\Delta_q}$. Taking invariants is exact in characteristic zero; hence this equality is an equality of the extensions inside the common punctured-disc bundle, not only an equality of their ranks or degrees. A translation twist induces the identity on the cohomology just used and changes none of these lattices. This proves the comparison at both finite points, including the full nonreduced fibres.

At the cusp the relative logarithmic forms have the global toric frame $d\log x_1, d\log x_2$, modulo $d\log q$. A toric gluing has the form $x_i \mapsto c_i(q)q^{b_i}x_i$ and therefore changes $d\log x_i$ by $(qc_i'/c_i + b_i)d\log q$. It induces the identity on the relative logarithmic quotient. The structure-sheaf comparison is also the identity on the normalization cochain complex. This verifies the cusp extension and the assertion about translated collars.

The logarithmic cotangent exact sequences are isomorphic on the smooth base, so their connecting maps agree there under these comparisons. Their source and target sheaves are locally free by the local calculations in Part A and the eigenframe calculation above. A holomorphic map between vector bundles on a reduced curve which is zero on a dense open set is zero everywhere. Thus the connecting maps agree globally. □

Statement B1.2 [Proved]. All admissible twists have the same Hodge numbers.

For every family in B1.1 whose cusp is the one-component hexagonal filling, the direct images of the absolute cotangent sheaves are

	$R^0 f'_*$	$R^1 f'_*$	$R^2 f'_*$
$\Omega^1_{X'}$	$\mathcal{O}(-2)$	$\mathcal{O} \oplus \mathcal{O}(-1)^2$	$\mathcal{O} \oplus \mathcal{O}(-1)$
$\Omega^2_{X'}$	$\mathcal{O}(-2) \oplus \mathcal{O}(-1)$	$\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$	$\mathcal{O}$

and $R^q f'_* \mathcal{O}_{X'} = (\mathcal{O}, \mathcal{O} \oplus \mathcal{O}(-1), \mathcal{O}(-1))$. In particular every admissible $(3,4)$ member with $|12l_0 - 4l_1 - 3l_2| = 1$ has the Hodge diamond of Part A.

Proof. B1.1 identifies the logarithmic direct images and their connecting maps. In Part A the cokernel of κ_0 is $\mathbb{C}_{p_1} \oplus \mathbb{C}_{p_2}$, and κ_1 is surjective with kernel $\mathcal{O} \oplus \mathcal{O}(-1)^2$. These statements therefore hold for X'. The residue sheaf of its normalized divisor has higher direct images

$$R^0 f'_* \mathcal{O}_{\widetilde{D}'} = \mathbb{C}_{p_0} \oplus \mathbb{C}_{p_1} \oplus \mathbb{C}_{p_2}, \quad R^1 f'_* \mathcal{O}_{\widetilde{D}'} = \mathbb{C}_{p_1} \oplus \mathbb{C}_{p_2}, \quad R^2 = 0.$$

Indeed the finite reductions are bielliptic, and the cusp normalization is rational. The base logarithmic residue is evaluation with weights $(1, m_1, m_2)$; its kernel is $\mathcal{O}(1)(-\Sigma) = \mathcal{O}(-2)$. On H^1 the residue of the invariant class $\gamma^{0,1}$ is $m_j \gamma^{0,1}|_{S'_j} \neq 0$. The other eigenline has zero residue. The local rank-one image of κ_0 is its kernel, as follows either from the residue diagram of Part A or from the eigenframe calculation in B1.1. Thus the two finite skyscrapers cancel exactly. This proves the first row, with no surviving torsion.

The proper flat map is Gorenstein of relative dimension two. Relative duality for $F = \Omega^1_{X'}$, using $F^* \otimes \omega_{X'/B} = \Omega^2_{X'} \otimes f'^* \mathcal{O}(2)$, gives the second row because all sheaves in the first row are locally free. Coherent Leray on $\mathbb{P}^1$ gives every Hodge number. In particular $h^{p,0} = 0$ for $p > 0$ follows from these direct images and duality, so no separate invariance assumption about absolute forms is needed. □

2.2 B.2. Higher multiplicities and the anticanonical ring

Statement B2.1 [Proved]. Canonical bundles for all coprime multiplicities.

Let $X = X_{3k,4l}(u)$ with admissible translation data as in Viaclovsky (2026b, Theorem 1.1 and Proposition 2.3), $\gcd(3k, 4l) = 1$, and any nonzero u in its construction domain. Then

$$K_X = f^* \mathcal{O}_B(-1) \otimes \mathcal{O}_X\big((k-1)S_1 + (3l-1)S_2\big).$$

Moreover $(h^{0,q})_{q=0}^3 = (1,1,0,0)$, $h^{p,0} = 0$ for $p > 0$, and $(h^{1,1}, h^{1,2}) = (2,1)$ for every such pair. Thus these Hodge numbers do not distinguish the multiplicity families.

Proof. The cited proposition supplies exactly the free cyclic quotients used in B1.1; its Theorem 1.1 supplies the stated sphere topology for its admissible choices. B1.2 computes the cohomology, with no use of that topological conclusion. For the canonical bundle we check the natural evaluation $f^* f_* \omega_{X/B} \to \omega_{X/B}$. The invariant relative determinant has weights $a_1 = 2k$ and $a_2 = l$ under the clockwise generators of orders $3k$ and $4l$. The pullback of a canonical form is $F(s)\, ds \wedge e$, where the character of $ds \wedge e$ is ζ_m^{1+a}. Therefore its invariant coefficients are s^{m-1-a} times an invariant function and a nowhere vanishing unit. Since the quotient of total spaces is etale, the evaluation vanishes to order $m - 1 - a$ on the reduced fibre downstairs. These orders are $k - 1$ and $3l - 1$. At the cusp the form $d \log x_1 \wedge d \log x_2 \wedge dq$ is a nowhere vanishing canonical frame in every smooth toric chart, so no cusp component occurs.

By B1.1 and relative duality, $f_* \omega_{X/B} = (R^2 f_* \mathcal{O}_X)^* = \mathcal{O}(1)$. The evaluation divisor just computed therefore gives $\omega_{X/B} = f^* \mathcal{O}(1) \otimes \mathcal{O}((k-1)S_1 + (3l-1)S_2)$. Tensoring by $f^* K_B = f^* \mathcal{O}(-2)$ proves the formula. □

Statement B2.2 [Proved]. The complete graded anticanonical ring.

Put $m_1 = 3k$, $m_2 = 4l$, and $d = 4l + 3k - kl$. Choose integers r, s with $m_2 r + m_1 s = 1$ and set $\mathcal{L} = \mathcal{O}_X(rS_1 + sS_2)$. There is a graded algebra isomorphism

$$\bigoplus_{n\geq 0} H^0(X, -nK_X) \simeq \bigoplus_{n\geq 0} \mathbb{C}[x,y]_{dn}, \qquad \deg x = m_2, \quad \deg y = m_1,$$

where the right-hand summand is assigned degree n. For negative weights the polynomial space is zero. Explicitly

$$h^0(X, -nK_X) = \max\left\{1 + n - \left\lceil \frac{n(k-1)}{3k} \right\rceil - \left\lceil \frac{n(3l-1)}{4l} \right\rceil, 0\right\}.$$

The integer d is nonzero for admissible pairs. If $d < 0$ the anticanonical ring is $\mathbb{C}$; if $d > 0$ it is the indicated weighted Veronese ring. For $(k,l) = (1,1)$ it is $\mathbb{C}[U,V]$, $\deg U = 1$, $\deg V = 2$.

Proof. The divisor of $f^*(t/(t-1))$ is $m_1 S_1 - m_2 S_2$. It follows, retaining this rational function as the comparison, that $\mathcal{L}^{m_2} \simeq \mathcal{O}(S_1)$, $\mathcal{L}^{m_1} \simeq \mathcal{O}(S_2)$ and $\mathcal{L}^{m_1 m_2} \simeq f^*\mathcal{O}(1)$. B2.1 then gives $-K_X = \mathcal{L}^d$.

We first establish the required section algebra for every integer power of $\mathcal{L}$. If a meromorphic function represents a section of $\mathcal{O}(aS_1 + bS_2)$, it is holomorphic on every regular compact fibre and is constant there. It descends to the base on the smooth locus. On a cyclic cover $q = s^m$, the finite pole bound gives a Laurent expansion bounded below, and invariance leaves only powers divisible by m. Thus descent extends meromorphically across the finite point, with the exact valuation inequality $m \operatorname{ord}_q(h) + a \geq 0$. The identical argument at the reduced cusp has $m = 1$. Conversely each base function satisfying these bounds pulls back to an allowed section. Consequently, as subsheaves of meromorphic functions,

$$f_*\mathcal{O}(aS_1 + bS_2) = \mathcal{O}_B\left(\lfloor a/m_1 \rfloor\, p_1 + \lfloor b/m_2 \rfloor\, p_2\right).$$

This proves the displayed dimension formula using B2.1 and $h^0(\mathbb{P}^1, \mathcal{O}(e)) = \max\{e+1, 0\}$.

For the algebra, let x, y be the canonical sections of $\mathcal{O}(S_1), \mathcal{O}(S_2)$ under the chosen identifications with powers of $\mathcal{L}$. Their powers x^{m_1}, y^{m_2} pull back two independent sections of $\mathcal{O}_B(1)$, with zeros at the two distinct points. For any integer j, choose the unique residues $0 \leq a_0 < m_1$, $0 \leq b_0 < m_2$ for which $j - m_2 a_0 - m_1 b_0 = e m_1 m_2$ is integral. The valuation formula identifies $H^0(\mathcal{L}^j)$ with $x^{a_0} y^{b_0} H^0(\mathcal{O}_B(e))$. When $e \geq 0$ a basis is

$$x^{a_0 + m_1 i} y^{b_0 + m_2(e-i)}, \qquad 0 \leq i \leq e;$$

when $e < 0$ there are no sections. These are exactly the monomials of weighted degree j. Their multiplication is ordinary polynomial multiplication, proving the algebra assertion in every degree. When $k = l = 1$, the weight-$6n$ monomials are generated by y^2 and x^3, of regraded degrees $1, 2$. Finally, $d = 0$ would imply $k \mid 4l$ and $l \mid 3k$; coprimality forces $k = l = 1$, which instead gives $d = 6$. □

For completeness the requested small pairs give the following exact data. Here r_1, r_2 are the coefficients in B2.1 and $P_n^- = h^0(-nK_X)$.

(k,l)	(m_1,m_2)	(r_1,r_2)	$(\deg x, \deg y)$	d	P_1^-	P_2^-	P_3^-
$(1,1)$	$(3,4)$	$(0,2)$	$(4,3)$	6	1	2	2
$(1,2)$	$(3,8)$	$(0,5)$	$(8,3)$	9	1	1	2
$(3,1)$	$(9,4)$	$(2,2)$	$(4,9)$	10	0	1	1
$(1,4)$	$(3,16)$	$(0,11)$	$(16,3)$	15	1	1	1
$(5,1)$	$(15,4)$	$(4,2)$	$(4,15)$	14	0	1	1
$(1,5)$	$(3,20)$	$(0,14)$	$(20,3)$	18	1	1	1

The program `b_families.py` enumerates all coprime pairs with $m_1 m_2 \le 60$ and checks the floor formula against exact weighted monomial counting for $0 \le n \le 12$. These checks accompany the all-degree proof above.

2.3 B.3. Finite fundamental groups and universal covers

Statement B3.1 [Proved]. Connected cyclic covers and the cusp comparison.

For an admissible $(3,4)$ member with $n = |12l_0 - 4l_1 - 3l_2| > 1$, let $\pi : Y \to X$ be the universal cover and $g = f\pi$. It is a finite etale holomorphic map of degree n. Every fibre of g is connected; its finite reductions are bielliptic, its multiplicities are $3, 4$, and its cusp W_n normalizes to a disjoint union of n copies of dP_6. The pullback maps induce isomorphisms

$$R^q f_* \mathcal{O}_X \simeq R^q g_* \mathcal{O}_Y, \qquad R^q f_* E_X \simeq R^q g_* E_Y.$$

Proof. We first derive the fundamental-group presentation from the fillings. The smooth family over the punctured base has zero section, so its fundamental group is $\Lambda \rtimes \langle x, y\rangle$, where x, y are the clockwise meridians and act by A_1, A_2. On a finite filling the lifted affine generator has m_jth power translation by v_j: its m_j translations add to v_j because $A_j v_j = v_j$. Thus filling the normal disc imposes $x^3 = t_{v_1}$ and $y^4 = t_{v_2}$, with the signs fixed by the clockwise convention on the first page.

The toric covering space of the cusp tube is simply connected. Indeed its dense torus has cocharacter lattice $\mathbb{Z}^3$; a loop may be perturbed into that torus, and the normal discs of the three rays $(0,1), (e_1,1), (e_2,1)$ kill a basis of this lattice. This argument applies also to the infinite fan, since a compact loop and a homotopy meet only finitely many of its locally finite charts. The same conclusion holds for the small tube: if a loop in $|t_c| < \epsilon$ bounds a disc in the toric space, the torus action $(x_1, x_2, t_c) \mapsto (x_1, x_2, a t_c)$ for a sufficiently small positive a moves that compact disc into the tube, and the scaling homotopy of its boundary stays in the tube. The cusp quotient therefore has fundamental group $\Lambda/\Lambda_{\mathrm{tor}}$, and its toric meridian bounds the normal disc at fixed x_1, x_2. A translated collar makes this meridian $t_\mu y^{-1} x^{-1}$, where $\gamma(\mu) = l_0$. Van Kampen consequently kills Λ_{tor} and imposes $xy = t_\mu$.

The normal closure of Λ_{tor} in the fibre lattice is $\ker\gamma$: it contains $\hat{w}, \hat{\delta}$ and $A_1\hat{w} + \hat{w} = \hat{u}$, and all these relations lie in $\ker\gamma$. The remaining generator c, the image of $\hat{\gamma}$, is central because $\gamma A_j = \gamma$. The presentation is thus $\langle c, x, y \mid c \text{ central},\ xy = c^{l_0},\ x^3 = c^{l_1},\ y^4 = c^{l_2}\rangle$. Eliminating y makes it abelian, with relation matrix

$$\begin{pmatrix} 3 & -l_1 \\ 4 & l_2 - 4l_0 \end{pmatrix}, \qquad \det = -p.$$

Its first Smith divisor is one because its entries include three and four; hence the group is $\mathbb{Z}/|p|$. After imposing $c = 1$ the quotient has $x^3 = x^4 = 1$, so it is trivial. Thus the image of the regular-fibre lattice, containing c, generates the entire group $\mathbb{Z}/n$. The same is true of each reduced special fibre. At a finite point the cyclic model is a smooth torus family over a disc, with a free affine finite-group action. An equivariant smooth trivialization, obtained by averaging a horizontal connection, contracts the disc to its centre and descends to a deformation retraction of the local total space onto S_j. Thus the image of the regular-fibre lattice in the local fundamental group is contained in the image of $\pi_1(S_j)$; the same holds after mapping to $\pi_1(X)$. At the cusp $\hat{\gamma}$ survives in $\Lambda/\Lambda_{\mathrm{tor}}$. The explicit normal-crossing contraction of the tube onto W given in B3.3 identifies their fundamental groups, so the same generator is in the image of $\pi_1(W)$. The covering-space criterion therefore makes the inverse image of every fibre connected. Admissibility gives $3 \nmid l_1$ and odd l_2, so $\gcd(n, 12) = 1$. The etale cover preserves

fibre multiplicities. On a finite reduction it cannot kill the nontrivial order-3 or order-4 linear holonomy, since the covering degree is prime to it; the resulting finite torus quotient is again bielliptic.

Normalization commutes with the finite etale base change $Y \to X$. The pullback of the simply connected normalization dP_6 is its trivial degree-n cover. The double curves and triple points also pull back etale, and their incidence maps are the lifts of the original incidence maps. Consequently the dual complex is an actual degree-n covering of the original real two-torus, rather than merely a cell complex with the same cell counts. It is connected because W_n is connected. The normalization resolution for $\mathcal{O}_{W_n}$ has acyclic components, double curves and point strata. Its global cochain complex is consequently the cellular cochain complex of this dual torus, with $n, 3n, 2n$ cells. Thus its coherent cohomology is $(1, 2, 1)$, and pullback from W is an isomorphism over $\mathbb{C}$: on a finite cover of real tori it is induced by a full-rank map of their integral first cohomology lattices.

Modulo $d \log q$, the two toric forms $d \log x_i$ are invariant under all cusp gluings as computed in B1.1. Hence $E_X|_W = \mathcal{O}_W^2$ and $E_Y|_{W_n} = \mathcal{O}_{W_n}^2$, and the same pullback argument applies to their cohomology. Over a regular fibre, an isogeny of complex tori induces isomorphisms on $H^q(\mathcal{O})$ and $H^q(\Omega^1)$ over $\mathbb{C}$. At a finite fibre, make the cyclic base change. The map is an isogeny of the smooth torus families and induces isomorphisms of their cohomology bundles. Taking the same cyclic invariant eigenspaces descends these isomorphisms, including all nilpotent layers of the scheme fibre. Grauert base change at the cusp, and this explicit calculation at the finite points, prove the asserted bundle maps are isomorphisms on all of B. □

Statement B3.2 [Proved]. Cotangent direct images and Hodge numbers of the covers.

Write $T_n = \mathbb{C}_{p_0}^{n-1}$. Then

	$R^0 g_*$	$R^1 g_*$	$R^2 g_*$
Ω_Y^1	$\mathcal{O}(-2)$	$\mathcal{O} \oplus \mathcal{O}(-1)^2 \oplus T_n$	$\mathcal{O} \oplus \mathcal{O}(-1)$
Ω_Y^2	$\mathcal{O}(-2) \oplus \mathcal{O}(-1)$	$\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$	$\mathcal{O} \oplus T_n$

The splittings involving T_n are not canonical. The only nonzero Hodge numbers of Y are

$$h^{0,0} = h^{0,1} = h^{3,2} = h^{3,3} = 1, \quad h^{1,2} = h^{2,1} = 1, \quad h^{1,1} = h^{2,2} = n + 1.$$

The original finite-fundamental-group threefold X itself has the same Hodge diamond as the simply connected $(3, 4)$ member.

Proof. B3.1 identifies the logarithmic cotangent sequence on cohomology, and therefore its connecting maps and direct images, with those of the original family. The normalized residue sheaf differs only at the cusp: its degree-zero cohomology there is now $\mathbb{C}^n$. The residue of a base logarithmic form is the diagonal vector $(1, \ldots, 1)$, since each cusp component has multiplicity one. Its cokernel is T_n. The degree-one residues at the finite fibres cancel the same two skyscrapers as in B1.2. The long exact sequence therefore gives

$$0 \longrightarrow T_n \longrightarrow R^1 g_* \Omega_Y^1 \longrightarrow \mathcal{O} \oplus \mathcal{O}(-1)^2 \longrightarrow 0$$

and the remaining entries of the first row. The extension splits because $\mathrm{Ext}^1(V, T_n) = H^1(V^* \otimes T_n) = 0$ for a vector bundle V on B.

For clarity relative duality must now be used in its derived form. For a sheaf F locally free on the total space, it gives

$$Rg_*(F^* \otimes \omega_{Y/B}) \simeq R\mathcal{H}om_B(Rg_* F, \mathcal{O}_B)[-2].$$

For $F = \Omega^1_Y$, the torsion in $R^1 g_* F$ contributes $\mathcal{E}xt^1_B(T_n, \mathcal{O}_B)$ in degree two of the right side. This is a length-$(n-1)$ skyscraper at p_0. After tensoring by $\mathcal{O}(-2)$ it still has that length. The locally free parts dualize in complementary degrees, giving exactly the second row; the extension by the locally free quotient splits for the same reason as above. Coherent Leray computes the displayed Hodge numbers. For X with one-component cusp, B1.2 already applies independently of its fundamental group. □

Statement B3.3 [Proved]. Canonical rings and rational topology of the covers.

For X and Y as above,

$$K_Y = \pi^* K_X = \mathcal{O}_Y(-2S_2'), \quad \bigoplus_{m \geq 0} H^0(Y, -mK_Y) = \mathbb{C}[U, V], \quad \deg(U, V) = (1, 2).$$

The same ring holds for X. Over $\mathbb{C}$, the Betti numbers of Y are $(1, 0, n-1, 0, n-1, 0, 1)$; its Frölicher sequence degenerates at E_2. For every nontrivial character of the deck group, the character part of Dolbeault cohomology consists of one class in $(1, 1)$ and one in $(2, 2)$ and no others.

Proof. The canonical formula pulls back through an etale map, and $4S_2' = g^* p_2$. B3.1 gives $g_* \mathcal{O}_Y = \mathcal{O}_B$. The valuation argument in B2.2 therefore applies unchanged and proves the ring statement with all multiplication maps.

The comparisons of B3.1 are pullback maps and identify the logarithmic cohomology bundles with the invariant part; the deck group acts trivially on those bundles. At the cusp its action on the n normalization components is the regular permutation representation. The quotient by the diagonal has each nontrivial character once. Thus the first-row additional skyscraper in B3.2 has precisely these characters, and duality gives the conjugate characters in its second-row additional skyscraper. All other cohomology is invariant and equal to that of X. Coherent Leray now proves the assertion about the Dolbeault character spaces.

The de Rham complex and its Frölicher sequence split under the idempotents of the finite deck group. A nontrivial character has just the two stated Dolbeault entries. No differential can connect them or connect either to another nonzero entry, so both survive. The invariant summand is the de Rham complex of X by finite etale descent and averaging. We compute its rational topology here from the actual attachment maps; the integral homology assertion of AC26 is not used as an input. In the remainder of this proof all homology and cohomology coefficients are $\mathbb{Q}$.

Let N be a small closed cusp tube, $M = \partial N$, and $U = X \setminus \operatorname{int} N$. The space U is covered by two pieces retracting onto S_1, S_2, whose intersection retracts onto a regular fibre F. The retractions are obtained from the equivariant smooth torus trivializations used in B3.1. The maps $F \to S_j$ are the central finite covering maps; translation of their origins induces the identity on their rational cohomology. Finite-cover transfer identifies their pullbacks with the invariant subspaces for A_j. Put

$$\psi_1 = 2u + w + 3\delta, \qquad \psi_2 = u + w + 2\delta, \qquad \eta = u \wedge w + 6\gamma \wedge \delta.$$

Direct substitution in A_1, A_2 shows that the invariant degree-one spaces are $\langle \gamma, \psi_j \rangle$ and the invariant degree-two spaces are $\langle \gamma \wedge \psi_j, \eta \rangle$. Their dimensions are two: the nontrivial eigenvalues of A_j on the complementary plane are the inverse primitive third or fourth roots, so the exterior-square fixed space has exactly the two displayed generators. In the transfer-normalized dual homology bases, the two Mayer–Vietoris maps are therefore the following actual matrices:

$$\alpha_1 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 1 & 3 \\ -1 & 0 & 0 & 0 \\ 0 & -1 & -1 & -2 \end{pmatrix}, \qquad \alpha_2 = \begin{pmatrix} 2 & 1 & 3 & 0 & 0 & 0 \\ 0 & 0 & 6 & 1 & 0 & 0 \\ -1 & -1 & -2 & 0 & 0 & 0 \\ 0 & 0 & -6 & -1 & 0 & 0 \end{pmatrix}.$$

The second matrix uses the ordered wedge basis $(\hat\gamma\hat u, \hat\gamma\hat w, \hat\gamma\hat\delta, \hat u\hat w, \hat u\hat\delta, \hat w\hat\delta)$. Both matrices have rank three, and $\ker\alpha_1 = \mathbb{Q}\nu_0$, with $\nu_0 = -\hat u - \hat w + \hat\delta$. Consequently

$$H_1(U) = \mathbb{Q}, \qquad 0 \longrightarrow \operatorname{coker}\alpha_2 = \mathbb{Q} \longrightarrow H_2(U) \xrightarrow{\partial} \mathbb{Q}\nu_0 \longrightarrow 0. \tag{B.1}$$

The generator $\Theta \in H^1(U)$ is normalized by $\Theta|_F = \gamma$. The finite affine generators have $3\bar g_1 = l_1 c$, $4\bar g_2 = l_2 c$ in rational homology, where c is the image of $\hat\gamma$ and $\Theta(c) = 1$. For the toric meridian $a_0 = t_\mu y^{-1}x^{-1}$ this gives the explicit collar value

$$\Theta(j_* a_0) = l_0 - l_1/3 - l_2/4 = p/12. \tag{B.2}$$

We justify the comparison with N without assuming a collapse theorem from AC26. In each normal-crossing chart $t_c = z_1 \cdots z_r$, $1 \le r \le 3$, choose the real vector field whose z_i component is $-z_i|z_i|^{-2}/\sum_j |z_j|^{-2}$ off the divisor. It satisfies $\dot t_c = -t_c$ and has norm at most $C|t_c|^{1/r}$. A finite partition of unity preserves the first identity and gives a global field with norm at most $C'|t_c|^{1/3}$ after shrinking the tube. Extend it by zero on W. Its flow exists for all positive time, and its tail length is bounded by $3C'\epsilon^{1/3}e^{-T/3}$ after time T. Thus the flow converges uniformly to W and gives a continuous strong deformation retraction fixing W. The same estimate proves continuity at the divisor and at the endpoint of the retraction.

The constant-sheaf normalization resolution of W has strata dP_6, three copies of $\mathbb{P}^1$, and two points. Its row of degree-zero cohomology is

$$\mathbb{Q} \xrightarrow{0} \mathbb{Q}^3 \xrightarrow{(a,b,c)\mapsto(a-b+c,\,a-b+c)} \mathbb{Q}^2.$$

The strata have no degree-one cohomology. This resolution therefore gives $H_1(N) = H_1(W) = \mathbb{Q}^2$, $H_4(N) = \mathbb{Q}[W]$ and $H_5(N) = 0$; it also gives $e(W) = 6 - 3\cdot 2 + 2 = 2$. Poincare–Lefschetz duality for the compact smooth manifold with boundary (N, M) now shows that $H_1(M) \to H_1(N)$ and $H_2(M) \to H_2(N)$ are surjective. For degree one the next relative group is $H_1(N, M) = H^5(N) = 0$. For degree two the map to $H_2(N, M) = H^4(N)$ is the intersection pairing with $[W]$, which is zero because W is the divisor of the global function t_c on N. The Wang sequence for the boundary mapping torus, with $\operatorname{rank}(M_0 - I) = 2$, gives $\dim H_1(M) = 3$. Its kernel to $H_1(N)$ is exactly $\mathbb{Q}a_0$: this meridian is nonzero under projection to the base circle and bounds its normal disc in N.

We exhibit two classes in $K_2 = \ker(H_2(M) \to H_2(N))$. For $\nu = \hat w, \hat\delta$, let ξ_ν be the two-torus obtained by translating the toric meridian around the angular circle S^1_ν. In the chart $t_c = z_0 z_1 z_2$ corresponding to the rays $(0,1), (e_1,1), (e_2,1)$, these are the boundaries of $D^2_{z_0} \times S^1_{z_i}$ with the other coordinate fixed and the radii chosen so that $|t_c| = \epsilon$ on the boundary. They therefore bound in N with the literal toric meridian; no choice of an unrecorded collar lift is being made. Cutting the meridian into the two arcs of the cover defining (B.1), the two cylinders have end circles differing by $(I - A_1^{-1})\nu$. With this orientation of the Mayer–Vietoris boundary, matrix multiplication gives

$$\partial j_* \xi_{\hat w} = -\nu_0, \qquad \partial j_* \xi_{\hat\delta} = 0. \tag{B.3}$$

The affine translation of either endpoint has no effect on its homology class, which explains why this boundary formula is the same for every translation twist.

Choose $\mathcal{A} \in H^2(U)$ restricting to η on F; it exists by Mayer–Vietoris because the two transfer-normalized classes η have equal restrictions. Fibrewise translation by $S^1_{\hat\delta}$ defines a global smooth action on U: $A_j\hat\delta = \hat\delta$, and all affine translations commute with this circle. Slant product of its action map with the circle defines $\iota : H^2(U) \to H^1(U)$. Naturality on F gives $\iota\mathcal{A}|_F = \iota_{\hat\delta}\eta = -6\gamma$. The restriction $H^1(U) \to H^1(F)$ is injective by (B.1), so $\iota\mathcal{A} = -6\Theta$. By the definition of slant product,

$$\langle \mathcal{A}, j_*\xi_{\hat\delta}\rangle = \langle \iota\mathcal{A}, j_* a_0\rangle = -p/2. \tag{B.4}$$

We may add the class pulled back from $(\mathbb{Q}\nu_0)^*$ under the boundary map in (B.1) so that $\langle \mathcal{A}, j_*\xi_{\hat{w}} \rangle = 0$; this does not change its restriction to F or (B.4). In the coordinates on $H_2(U)$ given by ∂/ν_0 and this normalized pairing, the two actual attachment cycles have matrix

$$\begin{bmatrix} j_*\xi_{\hat{w}} & j_*\xi_{\hat{\delta}} \end{bmatrix} = \begin{pmatrix} -1 & 0 \\ 0 & -p/2 \end{pmatrix}. \tag{B.5}$$

It is invertible over $\mathbb{Q}$ because admissibility implies $p \neq 0$. Thus the image of K_2 already fills $H_2(U)$. Likewise (B.2) says the image of $\ker(H_1(M) \to H_1(N))$ is nonzero in $H_1(U)$. Using the surjectivity to N just proved, Mayer–Vietoris for $X = U \cup N$ now gives $H_1(X) = H_2(X) = 0$. Every other fibre has Euler number zero, so additivity over the three special fibres and the smooth torus bundle gives $e(X) = 2$. Poincare duality then gives $b_4 = b_5 = 0$ and $b_3 = 0$. This proves the asserted rational homology of X for all admissible twists; the only changing determinant in (B.2)–(B.5) is a nonzero constant multiple of p. The integer matrices and all exterior-power identities were checked exactly in `code/b_topology.py`, with output `outputs/b_topology.txt`.

Part C's argument using this rational topology and the same Hodge table now gives degeneration of the invariant summand at E_2. Summing it with the nontrivial character spaces proves the Betti numbers and the degeneration statement for Y. In particular $e(Y) = 2n$, as also follows from the covering degree. □

3 Frölicher, Bott–Chern, Aeppli, and Hermitian metrics

3.1 C.1. The differentials and their dependence on the Hodge number

In this section X is the standard marked $(3,4)$ threefold and $h = h^{1,2}(X) = 1$ by Part A. We retain h in the formal calculations to specify exactly which conclusions depend on its evaluated value. Every assertion involving h is valid for every value consistent with the Hodge table

$q\backslash p$	0	1	2	3
3	0	0	0	1
2	0	h	$h+1$	1
1	1	$h+1$	h	0
0	1	0	0	0.

(9)

Thus no conclusion below silently uses $h = 1$. The geometric inputs for this part are the structure-sheaf calculation, the vanishing of holomorphic forms, and the sphere topology, which give (9); Statements A1.3 and A3.1 determine h. We use $d = \partial + \bar{\partial}$ on complex-valued smooth forms and the complex orientation for integration.

Statement C1.1 [Proved]. The complete Frölicher differential in terms of h.

For a compact complex threefold with (9) and the complex de Rham cohomology of S^6, the Frölicher spectral sequence satisfies $E_1 \neq E_2 = E_\infty$. Its only nonzero d_1 ranks are

$$\operatorname{rank} d_1^{0,1} = 1, \quad \operatorname{rank} d_1^{1,1} = h, \quad \operatorname{rank} d_1^{1,2} = h, \quad \operatorname{rank} d_1^{2,2} = 1.$$

In particular its two middle rows are exact.

Proof. Here $E_1^{p,q} = H^q(X, \Omega_X^p)$ and $d_r : E_r^{p,q} \longrightarrow E_r^{p+r,q-r+1}$ are complex-linear. For $r = 2$, any source in the two middle rows has target in one of $(2,0), (3,0), (3,1)$ or outside the table, except the source $(0,2)$, which is zero. Sources in row $q = 3$ outside $(3,3)$ are zero, and $(3,3)$ has no target. For $r = 3$ the only possible targets inside the table have sources $(0,2)$ or

$(0,3)$, both zero. For $r>3$ there is no arrow inside the table. A zero term remains zero on subsequent pages. Hence $E_2=E_\infty$.

The filtration on $H^k_{\mathrm{dR}}(X;\mathbb{C})$ has no nonzero associated graded term for $1\le k\le 5$. Every middle term of E_2 therefore vanishes. Consequently

$$0\longrightarrow\mathbb{C}\longrightarrow\mathbb{C}^{h+1}\longrightarrow\mathbb{C}^{h}\longrightarrow 0,\qquad 0\longrightarrow\mathbb{C}^{h}\longrightarrow\mathbb{C}^{h+1}\longrightarrow\mathbb{C}\longrightarrow 0$$

are exact, giving the stated ranks. The first rank is one, so E_1 cannot equal E_2. □

Statement C1.2 [Proved]. A basis on which all first differentials are specified.

Choose a nonzero $g\in H^{0,1}_{\bar\partial}(X)$, a basis $q_1,\dots,q_h$ of a complement to $\mathbb{C}d_1g$ in $H^{1,1}_{\bar\partial}(X)$, a basis $v_1,\dots,v_h$ of $H^{1,2}_{\bar\partial}(X)$, and a class e complementary to the image of $d_1:H^{1,2}\to H^{2,2}$. With $a=d_1g$, $c_j=d_1q_j$, $d_j=d_1v_j$, and $t=d_1e$, the first differential is

$$g\mapsto a\mapsto 0,\quad q_j\mapsto c_j\mapsto 0,\qquad v_j\mapsto d_j\mapsto 0,\quad e\mapsto t\mapsto 0.$$

These are bases in the corresponding bidegrees. If $h=1$ and a class q restricts to the invariant fibre class η described below, we may take $q_1=q$, $v_1=gq$, and $e=q^2$, after scaling g if necessary. Statements A1.3 and A3.2 prove these hypotheses for X, so this geometric choice applies to the threefold in the question.

Proof. The first assertions follow from the injectivity and surjectivity proved in C1.1. For the last assertion write $\eta=u\wedge w+6\gamma\wedge\delta$ in the marked integral cohomology of a regular fibre. Its complex-oriented square has integral -12. Its product with the $(0,1)$ image of γ is nonzero: the explicit period calculation in C3.7 gives its nonzero pairing with $d\zeta_2$. Thus gq and q^2 have nonzero fibre restrictions. The images under d_1 restrict to zero on the fibre, because the fibre is a complex torus and its invariant representatives are d-closed. Therefore q complements d_1g, gq spans the one-dimensional $H^{1,2}$, and q^2 complements $d_1H^{1,2}$. The extension used in this last choice is the actual Leray lift in Statement A3.2. This argument uses that extension; it does not infer it from a calculation on the smooth fibre alone. □

3.2 C.2. A computation from the two definitions

Statement C2.1 [Proved]. Bott–Chern and Aeppli tables as functions of h.

With

$$H^{p,q}_{BC}=\frac{\ker\partial\cap\ker\bar\partial\subset A^{p,q}}{\operatorname{im}\partial\bar\partial},\qquad H^{p,q}_{A}=\frac{\ker\partial\bar\partial\subset A^{p,q}}{\operatorname{im}\partial+\operatorname{im}\bar\partial},$$

the complete tables are

$q\backslash p$	0	1	2	3
3	0	0	1	1
2	0	h	$2h$	1
1	0	2	h	0
0	1	0	0	0

BC

$q\backslash p$	0	1	2	3
3	0	0	0	1
2	0	h	2	0
1	1	$2h$	h	0
0	1	1	0	0.

A (10)

In particular, substitution of the value $h=1$ proved in A3.1 gives the workbench's numerical tables. The totals are $(1,0,2,2,2,2,1)$ and $(1,2,2,2,2,0,1)$, respectively.

Proof. We apply the structure theorem for bounded double complexes over a field, Stelzig (2021, Theorem 3), to the actual algebraic double complex $(A^{\bullet,\bullet}(X), \partial, \bar{\partial})$. The word bounded refers to the bidegrees, not to the dimensions of the spaces of smooth forms, so that theorem applies here. It gives a direct-sum decomposition by isomorphisms of double complexes into squares and zigzags. An odd zigzag has one-dimensional total cohomology. All odd zigzags in intermediate total degree are therefore excluded by $b_1 = \cdots = b_5 = 0$. An odd zigzag of length greater than one would have at least two vertices of its cohomological total degree. The extreme diagonals $p + q = 0$ and $p + q = 6$ contain just one bidegree each, so the remaining odd zigzags are exactly one dot at $(0, 0)$ and one at $(3, 3)$.

An even zigzag of length $2r$ contributes a nonzero d_r to one of the two Frölicher spectral sequences. C1.1 and complex conjugation exclude $r \geq 2$ in both sequences. Thus the remaining summands are arrows. The horizontal arrows, with their multiplicities, are

source	target	multiplicity
$(0,1)$	$(1,1)$	1
$(1,1)$	$(2,1)$	h
$(1,2)$	$(2,2)$	h
$(2,2)$	$(3,2)$	1.

(11)

The vertical arrows are their reflections in $p = q$, with the same multiplicities. The correspondence with spectral-sequence ranks is Stelzig (2021, Proposition 6(1)); alternatively one computes column cohomology on each horizontal arrow directly.

We now evaluate the displayed definitions, rather than use a formula for the dimensions. On an arrow, the source is killed by neither the outgoing differential nor its total differential, whereas its target is closed under both. There is no composite $\partial\bar{\partial}$. Hence the target contributes one dimension to Bott–Chern cohomology, and the source contributes one dimension to Aeppli cohomology. On a square, only the top right corner is closed under both operators, and it is the image of $\partial\bar{\partial}$ from the bottom left corner. The other three corners of the square are images of ∂ or $\bar{\partial}$, while the bottom left corner is not in $\ker \partial\bar{\partial}$. Thus a square contributes to neither cohomology. A dot contributes once to each. These computations commute with direct sums, including the possibly infinite sum of squares. Counting the sources and targets in (11) and its reflection gives (10).

The program `code/c_double_complex.py` performs a separate kernel-and-image computation over $\mathbb{Q}$ and a source-and-target count. It includes an acyclic square with its signs, checks $\partial\bar{\partial} + \bar{\partial}\partial = 0$, and verifies the two answers for $h = 0, 1, 2, 3, 4, 5$. Its output is `outputs/c_double_complex.txt`. This is exact linear algebra, not a floating-point check and not a determination of h. □

Statement C2.2 [Proved]. Comparison with McHugh and Angella.

The specialization of McHugh's Theorem 6.3, Proposition 6.4, equation (7), and Section 7.1 to (9) agrees with C2.1. The corresponding relations and Figure 5 in Angella's Section 4 also agree. No correction to these specific formulas is required.

Proof. McHugh's quotient $K^{p,q}$ is zero in intermediate total degrees on every arrow and square just used: a closed corner of an arrow is the image of the incoming differential and a closed corner of a square is in both images. Thus $k^{p,q} = 0$ for $0 < p + q < 6$. Also $E_2^{3,1} = 0$ by C1.1 and $H^{1,1}(X; \mathbb{R}) = 0$ because $H^2(X; \mathbb{R}) = 0$. Reading the formulas in McHugh (2018, Theorem 6.3, Proposition 6.4, (7), pp. 20–25) therefore gives

$$h^{1,1}_{BC} = 2h^{0,1} = 2, \qquad h^{1,2}_{BC} = h^{1,2} = h, \qquad h^{2,2}_{BC} = -1 + (h + 1) + h = 2h.$$

The remaining entries in his table are precisely the boundary entries of (10). In Angella (2018, Section 4, pp. 13–16), the relation

$$h^{2,2}_{BC} = 2h^{2,1}_{BC} - 2h^{0,1} + 2$$

gives $2h$, and the boundary values reduce to those above because $h^{2,0} = h^{0,2} = 0$. This comparison concerns exactly the formulas used here; it does not certify every assertion in either article. □

3.3 C.3. The algebra, line bundles, and metric consequences

Statement C3.1 [Proved]. The real Bott–Chern Chern-class map and a global G.

Let G be a $\bar\partial$-closed smooth $(0,1)$ representative of a nonzero $g \in H^1(X, \mathcal{O}_X)$. Set

$$\mathsf{A} = [\partial G]_{BC}, \qquad \mathsf{B} = [\bar\partial \bar G]_{BC}.$$

These classes form a complex basis of $H^{1,1}_{BC}(X)$. With the exponential $z \mapsto e^{2\pi i z}$ and the Chern convention $c_{1,BC} = [iF/(2\pi)]$, the map of real Lie groups is

$$\mathrm{Pic}(X) = \{\exp(zg) : z \in \mathbb{C}\} \longrightarrow H^{1,1}_{BC}(X), \qquad \exp(zg) \longmapsto z\mathsf{A} + \bar z\mathsf{B}. \tag{12}$$

It is injective and its image is the real subspace fixed by conjugation.

There is an operator-explicit global representative. Normalize the Fubini–Study form by $\int_{\mathbb{P}^1} \omega_{FS} = 1$, put $\theta = f^*\omega_{FS}$, choose any smooth Riemannian metric on X, and let $\mathcal{G}_d$ be its Green operator for the real de Rham Laplacian. Then

$$\alpha = d^*\mathcal{G}_d\theta, \qquad G_0 = \alpha^{0,1} \tag{13}$$

satisfy $d\alpha = \theta$, $\bar\partial G_0 = 0$, $[G_0] \neq 0$, and $\exp([G_0]) = f^*\mathcal{O}_{\mathbb{P}^1}(1)$.

Proof. C1.1 shows that $[\partial G]_{\bar\partial} = d_1 g \neq 0$. The conjugate class B is nonzero, and its Dolbeault image is zero. Thus A, B are independent. C2.1 gives their completeness as a basis.

The exponential sequence and $H^1(X;\mathbb{Z}) = H^2(X;\mathbb{Z}) = 0$ identify $\mathrm{Pic}(X)$ with $H^1(X, \mathcal{O}_X) = \mathbb{C}g$, with no period lattice. On the smooth trivial line with unit frame s, define $\bar\partial_z s = -2\pi i z G \otimes s$. This is integrable. If $G = \bar\partial\phi_i$ locally, the holomorphic frames $e_i = e^{2\pi i z\phi_i}s$ have transition functions $e_j = e^{2\pi i z(\phi_j - \phi_i)}e_i$, which represent $\exp(zg)$. The metric-compatible Chern connection is

$$\nabla_z = d - 2\pi i(zG + \bar z\bar G), \qquad F_z = -2\pi i(z\partial G + \bar z\bar\partial\bar G).$$

This proves (12), including its sign. The basis proves injectivity and the assertion about its real image.

The form θ is closed and exact because $H^2(X;\mathbb{R}) = 0$. Hodge decomposition, together with $d\theta = 0$ and the commutation of d with the Green operator, gives $dd^*\mathcal{G}_d\theta = \theta$. The operator preserves real forms. Since $d\alpha$ has type $(1,1)$, its $(0,2)$ component says $\bar\partial G_0 = 0$. If $G_0 = \bar\partial u$, then $\theta = \partial\bar\partial(u - \bar u)$. A real multiple of $\mathrm{Im}\, u$ would be a global plurisubharmonic function, because θ is semipositive. The maximum principle on compact X makes this function constant. This contradicts the fact that θ is nonzero over the regular locus of f.

Finally (12) gives $c_{1,BC}(\exp([G_0])) = [\theta] = c_{1,BC}(f^*\mathcal{O}(1))$. Injectivity gives the asserted line-bundle equality. Formula (13) is a global formula involving a specified elliptic inverse; it is not a finite coordinate formula on the toric and cyclic charts, and it depends on the auxiliary metric. □

Statement C3.2 [Proved]. The factor 12 with a fixed collar orientation.

Use the marked cohomology class γ dual to $\hat{\gamma}$, the clockwise meridians and the lift relation of [AC26, Section 7.5], and put $p = 12l_0 - 4l_1 - 3l_2 = \pm 1$. Let g_γ be the unique coherent Leray class whose fibre image is $\gamma^{0,1}$. Then

$$\exp(12g_\gamma) = f^*\mathcal{O}_{\mathbb{P}^1}(-p). \tag{14}$$

For the standard data $(l_0, l_1, l_2) = (0, 1, -1)$, this is $\exp(12g_\gamma) = f^*\mathcal{O}(1)$. For the opposite value $p = 1$, the exponent representing $f^*\mathcal{O}(1)$ is $-12g_\gamma$.

Proof. We include the comparison that fixes the sign. A degree-one integral class $k\gamma$ on a regular fibre extends over the cyclic filling of order m_j precisely when the affine generator $\hat{g}_j$ can be assigned an integer value n_j satisfying $m_j n_j = k\gamma(v_j) = kl_j$. Admissibility gives $\gcd(l_j, m_j) = 1$, so this is precisely $m_j \mid k$. At the cusp γ extends, as follows from its actual specialization to $H^1(W; \mathbb{Z})$ in [AC26, Theorem B.1 and Proposition 7.12]. The global monodromy invariants have rank one generated by γ. It follows, by intersection of these three local images in the marked local system, that $H^0(B, R^1 f_* \mathbb{Z}) = 12\mathbb{Z}\gamma$. This recovers the degree-one part of [AC26, Proposition 7.26], without identifying the reduced and non-reduced fibre: this is topological cohomology, unchanged by nilpotents, whereas the coherent comparison uses the full fibre.

Let $U = \bigsqcup_i D_i$ be small discs around the special points and let V be their pair-of-pants complement with collars. Lift 12γ to integral classes Θ_i on $f^{-1}(D_i)$ and Θ_V on $f^{-1}(V)$. Existence over V follows from $H^2(V; \mathbb{Z}) = 0$ and the low-degree Leray sequence; over each disc it follows from the local images just computed. Write $a_j = \Theta_V(\hat{\rho}_j)$. On the two finite clockwise meridians the differences $\Theta_i - \Theta_V$ have periods

$$n_1 = 4l_1 - a_1, \qquad n_2 = 3l_2 - a_2.$$

At the cusp, the specified lift relation $\widetilde{a}_0 = t_\mu \hat{\rho}_2^{-1} \hat{\rho}_1^{-1}$, with $\gamma(\mu) = l_0$, and the null-homotopy of $\widetilde{a}_0$ in the cusp filling give $n_0 = -12l_0 + a_1 + a_2$. Hence

$$n_0 + n_1 + n_2 = -p. \tag{15}$$

These are the explicitly oriented boundary evaluations in [AC26, Proposition 7.27, pp. 62–63], not an unspecified sign of a spectral sequence differential.

Represent the lifts by real closed one-forms. On a collar their difference is the pullback of a real closed base form b_i with clockwise period n_i, modulo an exact form. This follows from the degree-one Leray sequence over the annulus, since the difference has zero fibre restriction and the annulus has no second cohomology. Subtract exact terms on smaller collars, extending their primitives with cutoffs, so the equality of forms holds wherever a partition of unity changes. For a real base cutoff ρ_i equal to one toward D_i and zero toward V, the resulting glued real one-form α on X has

$$d\alpha = f^*\vartheta, \qquad \vartheta = \sum_i d\rho_i \wedge b_i.$$

The form ϑ is a smooth real base two-form; thus $\bar{\partial}\alpha^{0,1} = 0$. Its fibre restriction is the $(0,1)$ image of 12γ, so the coherent Leray edge isomorphism $H^1(X, \mathcal{O}_X) \simeq H^0(B, R^1 f_* \mathcal{O}_X)$ identifies $[\alpha^{0,1}] = 12g_\gamma$.

In a positively oriented polar coordinate (r, φ) on D_i, $b_i = -n_i\, d\varphi/(2\pi)$ modulo an exact form. Because ρ_i decreases from one to zero as r increases,

$$\int_{D_i \cap V} d\rho_i \wedge b_i = -n_i \int_0^1 d\rho_i = n_i.$$

The exact part does not change this integral, since $d\rho_i$ is supported inside the collar. Therefore $\int_B \vartheta = -p$ by (15). C3.1 identifies the Bott–Chern class of $\exp(12g_\gamma)$ with $[f^*\vartheta]$. On $\mathbb{P}^1$ two real $(1,1)$ forms of the same integral differ by $i\partial\bar\partial$ of a real function (solve the ordinary scalar Laplace equation). Hence this class is the Chern class of $f^*\mathcal{O}(-p)$, and injectivity in C3.1 proves (14). The program `code/c_sign_and_fibre.py` verifies the integer intersection, (15), and the collar integral exactly. □

Statement C3.3 [Proved]. The complete Bott–Chern algebra, for every h.

The positive-total-degree ideal $N \subset H^{\bullet,\bullet}_{BC}(X)$ satisfies

$$H^{\bullet,\bullet}_{BC}(X) = \mathbb{C}\cdot 1 \oplus N, \qquad N^2 = 0.$$

Its nonzero bidegrees and dimensions are those of C2.1. Thus the square-zero conclusion is independent of the disputed value of h.

Proof. Take global $\bar\partial$-closed representatives G, Q_j, V_j, E of the classes g, q_j, v_j, e of C1.2. By the arrow computation in C2.1, bases for the positive Bott–Chern groups consist of

$$\partial G,\ \bar\partial\bar G; \qquad \partial Q_j,\ \bar\partial\bar Q_j; \qquad \partial V_j,\ \bar\partial\bar V_j; \qquad \partial E,\ \bar\partial\bar E; \qquad \nu,$$

where the pairs have bidegrees $(1,1), (2,1), (1,2), (2,2), (3,2), (2,3)$, respectively, and $\int_X \nu = 1$. In particular the natural map $H^{3,2}_{BC} \to H^{3,2}_{\bar\partial}$ is an isomorphism: on its unique horizontal arrow the closed target maps nontrivially to column cohomology.

Write $a = \partial G$, $b = \bar\partial\bar G$ and $c_j = \partial Q_j$. The graded Leibniz rule gives

$$\partial\bar\partial(G\wedge\bar G) = -a\wedge b, \quad \partial\bar\partial(G\wedge\bar Q_j) = -a\wedge\bar c_j, \quad \partial\bar\partial(\bar G\wedge Q_j) = b\wedge c_j.$$

Consequently $\mathsf{AB} = 0$, $\mathsf{A}\overline{\mathsf{C}_j} = 0$, and $\mathsf{BC}_j = 0$.

Let $f^*\mathcal{O}(1) = \exp(\lambda g)$, where $\lambda \neq 0$. Its curvature may be chosen to be the pullback of a two-form on B, whose square vanishes pointwise. By C3.1, $\lambda^2\mathsf{A}^2 + \bar\lambda^2\mathsf{B}^2 = 0$. The Dolbeault image of B^2 is zero. Hence $[a^2]_{\bar\partial} = 0$. But $a^2 = \partial(G\wedge a)$, and C1.1 says that $d_1 : H^{1,2} \to H^{2,2}$ is injective. Thus $[G\wedge a]_{\bar\partial} = 0$. Choose a smooth $(1,1)$ form Φ satisfying $G\wedge a = \bar\partial\Phi$. Applying ∂ proves $a^2 = \partial\bar\partial\Phi$. Conjugation gives $\mathsf{B}^2 = 0$ as well.

To treat AC_j, pair its Dolbeault image with the generator $g \in H^{0,1}_{\bar\partial}$. Since $\bar\partial c_j = 0$ and $G\wedge a = \bar\partial\Phi$,

$$\int_X a\wedge c_j\wedge G = -\int_X \bar\partial(\Phi\wedge c_j) = 0.$$

Dolbeault–Serre duality and the isomorphism $H^{3,2}_{BC} \to H^{3,2}_{\bar\partial}$ give $\mathsf{AC}_j = 0$. Complex conjugation gives $\mathsf{B}\overline{\mathsf{C}_j} = 0$.

Every remaining product whose bidegree lies in $[0,3]^2$ lands in $(3,3)$: it is a product of a $(1,1)$ class with a $(2,2)$ class, or of a $(2,1)$ class with a $(1,2)$ class. Each non-top generator displayed above is d-exact, since its chosen primitive is $\bar\partial$-closed, or is the conjugate of such a form. The products therefore have integral zero. Integration is an isomorphism $H^{3,3}_{BC} \to \mathbb{C}$ by C2.1 and the class of ν, so all these products vanish. All other products exceed the complex dimension in at least one bidegree. This exhausts the possibilities. □

Statement C3.4 [Proved]. Non-Kähler degrees and the precise degreewise statement.

For $k = 0, \dots, 6$ the non-Kähler degrees are

$$(\Delta^k) = (0, 2, 2 + 2h, 4h, 2 + 2h, 2, 0). \tag{16}$$

If $h \geq 1$, the two maps $\bigoplus_{p+q=k} H^{p,q}_{BC} \to H^k_{\mathrm{dR}}(X; \mathbb{C}) \to \bigoplus_{p+q=k} H^{p,q}_A$ are both isomorphisms exactly for $k = 0, 6$. For the weaker assertion that a d-exact pure d-closed form of degree k is $\partial\bar{\partial}$-exact, the valid degrees are $k = 0, 1, 6$, and also $k = 3, 4$ if $h = 0$.

Proof. The totals from C2.1 are

$$(h^k_{BC}) = (1, 0, 2, 2h, 2h, 2, 1), \qquad (h^k_A) = (1, 2, 2h, 2h, 2, 0, 1).$$

Subtracting twice the Betti numbers gives (16). On the two endpoint dots both comparison maps are isomorphisms; all intermediate de Rham groups vanish. The first assertion follows by inspecting when both outside groups vanish. The weaker assertion is exactly injectivity of the first comparison map. In intermediate degree this is equivalent to vanishing of the Bott–Chern group alone, which gives the stated list. In degree zero there is no nonzero exact form, and in degree six integration identifies Bott–Chern and de Rham cohomology. This distinguishes the two meanings of a "$\partial\bar{\partial}$-lemma in a degree"; a vanishing Bott–Chern group in degree one does not make Δ^1 vanish. □

Statement C3.5 [Proved]. Metric obstructions that do not require h.

The threefold X admits no balanced, strongly Gauduchon, Kähler, locally conformally Kähler, or Hermitian-symplectic metric.

Proof. Let $\theta = f^*\omega_{FS}$. It is a nonzero semipositive exact real $(1,1)$ form. If ω were balanced, the real four-form ω^2 would be closed and $\int_X \theta \wedge \omega^2 > 0$, because the integrand is positive on the regular locus. Exactness of θ and Stokes' theorem give zero, a contradiction.

More generally, if ω were strongly Gauduchon, write $\partial\omega^2 = \bar{\partial}u$ with $u \in A^{3,1}$. The real four-form $\Omega = \omega^2 - u - \bar{u}$ is closed. Bidegrees give $\theta \wedge \Omega = \theta \wedge \omega^2$, so the same argument applies. Kähler metrics are balanced. If $d\omega = \ell \wedge \omega$ with a closed real Lee form ℓ, then $b_1 = 0$ gives $\ell = dv$, and $d(e^{-v}\omega) = 0$, excluding locally conformally Kähler metrics. Finally any symplectic two-form on X is exact because $b_2 = 0$. Its cube has nonzero integral for its symplectic orientation, whereas an exact cube has zero integral. Thus X is not symplectic and in particular admits no Hermitian-symplectic metric. □

Statement C3.6 [Known]. Gauduchon metrics with specified Chern–Ricci form.

There exists a Gauduchon metric on X. More precisely, for every Gauduchon metric ω_0 there is another Gauduchon metric ω with $[\omega^2]_A = [\omega_0^2]_A$ and

$$\mathrm{Ric}^{\mathrm{Ch}}(\omega) = \pi f^*\omega_{FS}. \tag{17}$$

This Ricci form is semipositive and has rank one at regular points of f. No Chern–Ricci-flat Hermitian metric exists on X.

Proof. Gauduchon's conformal theorem (Gauduchon, 1985, Theorem 2, pp. 132–133) applies to the compact connected complex manifold X. In complex dimension three, it produces a positive conformal multiple ω_0 of any Hermitian form with $\partial\bar{\partial}\omega_0^2 = 0$. For clarity, its reduction is a scalar elliptic equation: if $L\phi$ is a nonzero fixed normalization of $\mathrm{tr}_\omega(i\partial\bar{\partial}\phi)$, then L has

kernel the constants by the maximum principle and Fredholm index zero. The one-dimensional kernel of L^* has a strictly positive generator u. Indeed a sign-changing generator would admit a nonzero nonnegative function orthogonal to it, placing such a function in $\operatorname{im} L$, contrary to the maximum principle; the strong maximum principle then gives strict positivity. The form $u^{1/2}\omega$ is Gauduchon.

The canonical-bundle formula in Part A and $4S_2 = f^*(p_2)$ give $K_X^2 = f^*\mathcal{O}(-1)$. Consequently $c_{1,BC}(T_X) = \frac{1}{2}[\theta]$ with $\theta = f^*\omega_{FS}$. We use the conventional unnormalized Chern–Ricci form $-i\partial\bar\partial \log\det g$, which represents $2\pi c_{1,BC}(T_X)$. Thus the real closed form $\pi\theta$ represents its class. Applying Székelyhidi et al. (2017, Theorem 1.3) with background ω_0 and prescribed form $\pi\theta$ proves (17) and preservation of the Aeppli class. Finally C3.1 shows $[\theta]_{BC} \neq 0$. A metric with zero Chern–Ricci form would make $c_{1,BC}(T_X) = 0$, which is impossible. □

Statement C3.7 [Proved]. The fibre obstruction to SKT and astheno-Kähler metrics.

The threefold X admits no pluriclosed (SKT) metric and no astheno-Kähler metric. The geometric input is the value $h = 1$ and the global Dolbeault class q in A3.2, whose restriction to a regular fibre is $\eta = u \wedge w + 6\gamma \wedge \delta$. More generally, it is enough that the real image of $H_A^{1,1}(X;\mathbb{R}) \to H_A^{1,1}(F;\mathbb{R})$ be contained in $\mathbb{R}\eta$ for one regular fibre F.

Proof. We first compute the sign that matters. Put $y = \operatorname{Im}\tau > 0$, $m = \operatorname{Im}\mu$, and $\Delta = \operatorname{Im}\beta - 6m^2/y < 0$. For the marked period rows $\sigma_1 = 6\mu\gamma + \tau u + w$ and $\sigma_2 = \beta\gamma + \mu u + \delta$ we have $\eta = u \wedge \sigma_1 + 6\gamma \wedge \sigma_2$. This proves that the real class η has type $(1,1)$. The determinant of the four real period columns, with rows ordered as $\operatorname{Re}\zeta_1, \operatorname{Im}\zeta_1, \operatorname{Re}\zeta_2, \operatorname{Im}\zeta_2$, is $y\Delta < 0$. Since $\eta^2 = 12\gamma \wedge u \wedge w \wedge \delta$,

$$\int_{F,\mathrm{or}_{\mathbb{C}}} \eta^2 = -12. \tag{18}$$

Equivalently the invariant representative of η has Hermitian coefficient matrix

$$\frac{1}{y\Delta}\begin{pmatrix} \operatorname{Im}\beta & -6m \\ -6m & 6y \end{pmatrix}, \qquad \det = \frac{6}{y\Delta} < 0,$$

so it has one positive and one negative eigenvalue. The exact symbolic determinant and an exact rational example are checked in `code/c_sign_and_fibre.py` by both eigenvalues and diagonal elimination.

The explicit invariant fibre representative of $\gamma^{0,1}$ is

$$\Gamma = \frac{i}{2\Delta}\left(d\bar\zeta_2 - \frac{m}{y}d\bar\zeta_1\right).$$

Writing Q_F for the invariant $(1,1)$ representative of η, direct exterior multiplication gives

$$Q_F \wedge \Gamma = \frac{1}{4y\Delta} d\zeta_1 \wedge d\bar\zeta_1 \wedge d\bar\zeta_2, \quad \int_F Q_F \wedge \Gamma \wedge d\zeta_2 = -1.$$

This also proves the nonvanishing used in C1.2.

When $h = 1$, C2.1 identifies $H_A^{1,1}(X)$ with the sum of the one horizontal source and its vertical conjugate. A representative Q of q is nonzero in the horizontal source, since its fibre restriction is nonzero whereas $d_1 g$ restricts to zero. Thus $[Q]_A, [\bar Q]_A$ form a basis. Their restrictions to F both equal the real class η. The real image of the restriction map is therefore $\mathbb{R}\eta$.

If ω were a pluriclosed positive real $(1,1)$ form, its restriction to F would be pluriclosed and positive. Average this restriction under translations of the compact complex torus F.

The average ω_{av} is invariant, closed, and strictly positive. Translation preserves the Aeppli class: on a torus the $\partial\bar\partial$-lemma identifies Aeppli cohomology with the invariant (p,q) forms, so the average is the unique invariant representative of that class. This identification can also be checked Fourier mode by Fourier mode: each nonzero mode is a Koszul square for two independent type components of its real nonzero covector and contributes zero Aeppli cohomology. Thus $[\omega_{\mathrm{av}}] = r\eta$ for some real r. But positivity gives $\int_F \omega_{\mathrm{av}}^2 > 0$, whereas (18) gives $-12r^2 \le 0$, a contradiction. In dimension three the astheno-Kähler equation $\partial\bar\partial\omega^{n-2} = 0$ is exactly the SKT equation. □

Statement C3.8 [Open]. A literal global coefficient atlas for the generators.

The invariant computations and all the metric types named above are settled for X. The real restriction map

$$r_b : H_A^{1,1}(X;\mathbb{R}) \longrightarrow H_A^{1,1}(F_b;\mathbb{R})$$

has image $\mathbb{R}\eta$ by A3.2 and C3.7. Its source has real dimension two and its target has real dimension four. This concerns cohomology classes and does not assert literal equality of arbitrary smooth representatives with an invariant fibre form.

The formula (13) supplies a global G after a metric choice. For the standard primitive marking, $G_\gamma = G_0/12$ represents g_γ by C3.2. There is a precise operator construction of Q as well. Fix a Hermitian metric on X and a regular fibre F, identify η with its real invariant form on F, and choose an orthonormal basis e_1, e_2 of the Dolbeault-harmonic $(1,1)$ forms for $\Delta_{\bar\partial}$. Define

$$\ell(e) = -\frac{1}{12}\int_F i_F^* e \wedge \eta, \qquad Q = \frac{\overline{\ell(e_1)}e_1 + \overline{\ell(e_2)}e_2}{|\ell(e_1)|^2 + |\ell(e_2)|^2}.$$

The denominator is positive by A3.2 and $\int_F \eta^2 = -12$. The expression is independent of the orthonormal basis, is $\bar\partial$-closed, and has fibre Dolbeault class η. Indeed the image of fibre restriction on $H_{\bar\partial}^{1,1}(X)$ is $\mathbb{C}\eta$, with kernel $\mathbb{C}d_1 g$; the displayed functional measures exactly the coefficient of η, and the formula makes that coefficient one.

These two formulas fix global smooth representatives by explicit elliptic operators and a finite normalization. We have not evaluated their coefficient functions on a common atlas through the three special fibres. The remaining task is therefore precise: compute the harmonic projector and the two displayed restriction integrals for one fully specified global Hermitian metric, or construct and evaluate a Cech representative and partition-of-unity correction with the same restrictions. No numerical computation of those infinite-dimensional elliptic inverses was run. This is the unfulfilled literal-coefficient requirement, not an undecided Hodge number, product, or existence of the Leray class.

4 D. Tangent cohomology and deformations

We retain the standard filling data $(\ell_0, \ell_1, \ell_2) = (0, 1, -1)$ and an arbitrary admissible c. All sheaves in this section are analytic, and all cohomology groups have coefficients in $\mathbb{C}$. The integral twists below mean the actual line bundles

$$M_n = \mathcal{O}_X\big(n(S_1 - S_2)\big), \qquad n \in \mathbb{Z}.$$

We set $\mathfrak{g} = g_\gamma$. Thus, in the positively normalized exponential coordinate of C3.2, $M_n = \exp(n\mathfrak{g})$. The computations use the root base to retain the nonreduced fibres. Passing to its coarse space is a separate, exact invariant operation; it is never identified with taking the dual of an invariant lattice.

Statement D1.1 [Proved]. The root base and its explicit cohomology.

Let $\rho : \mathcal{B} \to B$ be the analytic root curve with orders 3 and 4 at p_1, p_2. Write R_1, R_2 for its tautological root line bundles. Its Picard group is infinite cyclic, with generator $R = R_1 \otimes R_2^{-1}$, and

$$R_1 = R^4, \quad R_2 = R^3, \quad \rho^*\mathcal{O}_B(1) = R^{12}, \quad K_{\mathcal{B}} = R^{-7}.$$

We abbreviate R^n to $\mathcal{O}_{\mathcal{B}}(n)$. Choose sections $x \in H^0(R^3)$ and $y \in H^0(R^4)$ vanishing at p_2 and p_1, respectively. For every integer n, put

$$r_1(n) \equiv n \pmod 3, \quad 0 \le r_1(n) < 3, \qquad r_2(n) \equiv 3n \pmod 4, \quad 0 \le r_2(n) < 4,$$

$$d(n) = \frac{n - 4r_1(n) - 3r_2(n)}{12}.$$

Then

$$\rho_*\mathcal{O}_{\mathcal{B}}(n) = \mathcal{O}_B(d(n)), \qquad H^i(\mathcal{B}, \mathcal{O}(n)) = H^i(B, \mathcal{O}(d(n))). \tag{D.1}$$

Equivalently, $h^0(\mathcal{O}(n))$ is the number of pairs $(a, b) \in \mathbb{Z}^2_{\ge 0}$ with $3a + 4b = n$, and $h^1(\mathcal{O}(n)) = h^0(\mathcal{O}(-n-7))$.

Proof. On the root charts the group acts on a disc coordinate by a primitive m_jth root of unity. The stabilizer character of any line bundle is removed uniquely by tensoring with $R_1^{-r_1}R_2^{-r_2}$, with the exponents in the displayed ranges. A line bundle with trivial stabilizer characters descends: an equivariant nonzero local frame can be averaged, and its invariant frame gives the descent transition function. It is therefore the pullback of a unique line bundle on B. Consequently

$$\mathrm{Pic}(\mathcal{B}) = \langle H, R_1, R_2 \mid 3R_1 = H,\ 4R_2 = H \rangle = \mathbb{Z}R.$$

Here and below additive notation for this presentation means tensor product. The determinant of the coordinate change $t = s^m$ gives $K_{\mathcal{B}} = \rho^*K_B \otimes R_1^2R_2^3 = R^{-24+8+9}$.

An invariant section of $R_j^{r_j}$ on its chart is s^{r_j} times a holomorphic function of $t = s^{m_j}$. It follows, by the local invariant comparison on overlaps, that $\rho_*(\rho^*\mathcal{O}(k)R_1^{r_1}R_2^{r_2}) = \mathcal{O}(k)$. In characteristic zero taking finite-group invariants is exact: the averaging idempotent splits the invariant submodule. Thus there are no higher direct images for ρ, and (D.1) follows. This proves the cohomology comparison in the direction used here, not merely equality of Euler characteristics. Finally every monomial of degree n is $x^{r_2}y^{r_1}$ times a homogeneous polynomial of degree $d(n)$ in x^4, y^3. This gives the stated count of sections. The negative-degree Cech complex on $x \neq 0$, $y \neq 0$ has cokernel represented by $x^{-a}y^{-b}$ with $a, b \ge 1$ and $3a + 4b = -n$; replacing (a, b) by $(a-1, b-1)$ proves the h^1 formula without an additional duality assumption. The two calculations agree exactly for every integer n; the program `code/d_exact.py` also compares them for $-240 \le n \le 240$. □

Statement D1.2 [Proved]. Vertical and logarithmic tangent bundles.

There is a canonical representable proper flat holomorphic map $\bar{f} : X \to \mathcal{B}$ with $f = \rho\bar{f}$. If

$$\mathcal{T} = T_X(-\log W), \qquad \mathcal{V} = \ker(\mathcal{T} \to \bar{f}^*T_{\mathcal{B}}(-p_0)),$$

then, for rank-two vector bundles E, G on $\mathcal{B}$,

$$\mathcal{V} = \bar{f}^*E, \quad E = \mathcal{O} \oplus \mathcal{O}(-1), \qquad G = R^1\bar{f}_*\mathcal{O}_X = \mathcal{O} \oplus \mathcal{O}(-1), \tag{D.2}$$

$$R^0\bar{f}_*\mathcal{O}_X = \mathcal{O}, \qquad R^2\bar{f}_*\mathcal{O}_X = \mathcal{O}(-1),$$

and the following is an exact sequence of locally free sheaves on X:

$$0 \longrightarrow \bar{f}^*E \longrightarrow \mathcal{T} \longrightarrow \bar{f}^*\mathcal{O}(-5) \longrightarrow 0. \tag{D.3}$$

The trivial summands in E, G are generated by the vertical field ξ and the constant integral cocycle γ, respectively.

Proof. The divisors S_j and their canonical sections give root data $\mathcal{O}_X(S_j)^{m_j} = f^*\mathcal{O}_B(p_j)$, hence the indicated map to the root curve. Over its order-m_j chart the fibre product is the original smooth torus family J'_j, since its free affine cyclic quotient is exactly $f^{-1}(D_j)$. The comparison $J'_j \to X$ is finite etale; in particular pullback is faithful and the stated local comparisons descend. Away from the two root points the map is the original f. Properness and flatness may be checked after this atlas: properness is the properness of the original fillings, and flatness holds because a nonconstant base parameter is a non-zero-divisor in every smooth total-space local ring. Fibres have dimension two, including the cusp.

On a smooth torus chart a vertical field is a holomorphic linear combination of the translation fields. Its coefficients are constant on each compact fibre and therefore depend only on the base. The vertical derivative of an affine transition is its linear part R_g; the translating part, including the logarithmic regluing, has vertical derivative the identity. At the cusp the two fields $x_1\partial_{x_1}, x_2\partial_{x_2}$ extend as logarithmic vector fields in every toric chart. The deck maps multiply each x_i by a base-dependent unit and a power of the base parameter. Their vertical logarithmic derivative fixes both coefficient directions. The exponential overlap sends ∂_{ζ_i} to $2\pi i x_i\partial_{x_i}$. These transitions depend only on the base and define E; they prove the actual bundle identity $\mathcal{V} = \bar{f}^*E$.

We check the kernel and surjectivity at every type of cusp point. At $t = xy$, the kernel of df is freely generated by $x\partial_x - y\partial_y$ and ∂_z. At $t = xyz$ it is generated by $x\partial_x - y\partial_y$ and $x\partial_x - z\partial_z$. Indeed the equation $yza + xzb + xyc = 0$ in the regular local ring forces $x \mid a, y \mid b, z \mid c$; after division the only relation is that the three coefficients sum to zero. The logarithmic map sends $\sum a_i z_i\partial_{z_i}$ to $(\sum a_i)t\partial_t$ and is surjective. At a finite root point it is the ordinary relative tangent sequence of the smooth map to s, not the singular differential of s^{m_j}. These comparisons prove (D.3), since $T_{\mathcal{B}}(-p_0) = \mathcal{O}(7-12)$.

The invariant vector e_2 defines a subbundle $\mathcal{O}\xi \subset E$. It remains nonzero as a vector of the logarithmic vertical bundle at the cusp, even where its image in the ordinary tangent fibre is zero. The quotient line has automorphy factors $-1/\tau$ and $1/\tau$ under the order-three and order-four generators. The modular discriminant has weight twelve and no zero in the upper half-plane. Therefore its reciprocal is a meromorphic section of the twelfth tensor power of this quotient line. On the cusp chart $\tau = (\log q)/(2\pi i) + h(q)$ with h holomorphic, so the product expansion of the discriminant is q times a holomorphic unit. There are no other zeros or poles, including on the finite root charts. The quotient line thus has twelfth power $\mathcal{O}(-p_0) = \mathcal{O}(-12)$, and the torsion-free Picard group in D1.1 makes it $\mathcal{O}(-1)$. Its extension by $\mathcal{O}$ splits because $H^1(\mathcal{B}, \mathcal{O}(1)) = 0$.

For completeness the structure-sheaf computation at the cusp uses the normalization $P = \mathrm{d}P_6$, the three double curves and the two triple points. Their exact normalization complex has global section complex

$$\mathbb{C} \xrightarrow{0} \mathbb{C}^3 \xrightarrow{(a,b,c)\mapsto(a-b+c,a-b+c)} \mathbb{C}^2.$$

Exactness at the two local models is the elementary gluing of functions on the coordinate branches; all higher cohomology of the rational normalization, the rational double curves and the points vanishes. The complex therefore gives $(h^0, h^1, h^2) = (1, 2, 1)$. Its product is the Cech product of the two-triangle torus and gives a nondegenerate alternating pairing $H^1 \wedge H^1 \to H^2$. This also follows directly from the two opposite oriented triangles: the two coordinate cocycles have determinant one on their oriented fundamental cycle. These are the local comparison and product in Claude and Alpoge (2026, Lemma 9.12), used here on the same normalization and with both triple points retained.

On all other geometric fibres of $\bar{f}$ the cohomology is that of a smooth two-torus. Constancy of these three dimensions and proper flat analytic base change give locally free direct images on the atlas. Their canonical comparison maps commute with the finite group actions and with overlap restriction; hence they descend to $\mathcal{B}$. The determinant of (D.3), using $W = \bar{f}^* p_0$, gives

$$\omega_{X/\mathcal{B}} = \bar{f}^*(\det E)^{-1}.$$

Relative Serre duality, applied on that same atlas and descended by its trace pairing, gives $R^2\bar{f}_*\mathcal{O}_X = \det E = \mathcal{O}(-1)$. Cup product is an isomorphism $\det G \to R^2\bar{f}_*\mathcal{O}_X$, since it is so on every geometric fibre by the preceding calculation.

The class γ is nowhere zero in G. On a smooth marked fibre, $H^1(\mathcal{O})$ is the quotient of constant lattice cocycles by the two period rows. If γ were a combination of those rows, evaluation on $\widehat{w}, \widehat{\delta}$ would make both coefficients zero, contradicting $\gamma(\widehat{\gamma}) = 1$. The same argument applies on each finite root cover, and affine translations act trivially on these classes. At the cusp γ vanishes on the angular sublattice and descends to a nonzero character of its rank-two deck lattice. In the normalization complex it is the corresponding nonzero one-cocycle on the two-triangle torus. Its cup product with a complementary integral character is nonzero, so it remains nonzero there. Thus $\mathcal{O}\gamma \subset G$ is a subbundle; its quotient is $\mathcal{O}(-1)$ by the determinant just computed. Once again $H^1(\mathcal{O}(1)) = 0$ splits the extension. This proves every assertion in (D.2). □

Statement D1.3 [Proved]. The untwisted tangent direct images.

For every admissible c in the standard family,

$$\boxed{f_*T_X = \mathcal{O} \oplus \mathcal{O}(-1), \quad R^1 f_* T_X = \mathcal{O} \oplus \mathcal{O}(-1)^2, \quad R^2 f_* T_X = \mathcal{O}.} \tag{D.4}$$

Consequently $(h^0, h^1, h^2, h^3)(X, T_X) = (1, 1, 1, 0)$.

Proof. Put $L = \mathcal{O}_{\mathcal{B}}(-5)$. The first connecting map of (D.3) is the holomorphic bundle map $\kappa : L \to E \otimes G$. In the splittings fixed in D1.2 the coefficient in $\mathcal{O}\xi \otimes \mathcal{O}\gamma$ lies in $H^0(\mathcal{O}(5)) = 0$. The remaining coefficients lie in $H^0(\mathcal{O}(4)) = \mathbb{C}y$, twice, and $H^0(\mathcal{O}(3)) = \mathbb{C}x$. Therefore, as a map from E^* to G with its L factor understood,

$$\kappa = \begin{pmatrix} 0 & ay \\ by & c_0 x \end{pmatrix}. \tag{D.5}$$

We use c_0 for this coefficient to distinguish it from the deformation parameter. The logarithmic period derivative at the cusp has leading term the invertible matrix $B_0/(2\pi i)$. Hence $\det \kappa = -aby^2$ is nonzero there, so $ab \neq 0$. At the order-three point $\tau - \rho$ has order one on the root chart: $j = 1728t$ and the elliptic ramification order of j at ρ is three. The marked torus period derivative is consequently nonzero there: the two constant identity columns in the normalized period matrix exclude an infinitesimal linear-frame change that could otherwise conceal a nonzero derivative. In (D.5), $y = 0$ and $x \neq 0$ at that point, so $c_0 \neq 0$. These arguments establish the three nonzero coefficients; they do not infer them from a global cohomology dimension. Rescaling the two quotient frames and the source frame makes all three coefficients one.

The next connecting map is cup product with κ. In suitable bases it is

$$L \otimes G = \mathcal{O}(-5) \oplus \mathcal{O}(-6) \xrightarrow{\delta_1} E \otimes \det G = \mathcal{O}(-1) \oplus \mathcal{O}(-2), \qquad \delta_1 = \begin{pmatrix} -y & 0 \\ -x & y \end{pmatrix}. \tag{D.6}$$

The sign follows by wedging first with γ and then with the second generator of G; reversing both sign conventions leaves the cokernels unchanged. As its determinant is $-y^2$, it is injective

as a morphism of sheaves. The first connecting map is $\delta_0 = (0, y, y, x)^t$ and is a subbundle inclusion, since x, y have no common zero. Its cokernel is

$$\mathcal{O} \oplus \mathcal{O}(-1) \oplus \mathcal{O}(2). \tag{D.7}$$

Indeed one constant linear combination of the two $\mathcal{O}(-1)$ summands is unused, and the remaining quotient is given by the exact sequence $0 \to \mathcal{O}(-5) \xrightarrow{(y,x)} \mathcal{O}(-1) \oplus \mathcal{O}(-2) \xrightarrow{(x,-y)} \mathcal{O}(2) \to 0$.

The cokernel of (D.6) is supported at the order-three root point. There x is a unit, and elementary row and column operations identify it with

$$Q = \mathcal{O}(-1)\big|_{2p_1}, \tag{D.8}$$

where $2p_1$ means the length-two divisor on the root chart, not the double coarse point. The generator is the first target component, and its annihilator is y^2. The long direct-image sequence now gives

$$R^0\bar{f}_*\mathcal{T} = E, \quad R^1\bar{f}_*\mathcal{T} = \mathcal{O} \oplus \mathcal{O}(-1) \oplus \mathcal{O}(2),$$
$$0 \longrightarrow Q \longrightarrow R^2\bar{f}_*\mathcal{T} \longrightarrow \mathcal{O}(-6) \longrightarrow 0. \tag{D.9}$$

This last extension splits, since $\mathrm{Ext}^1(\mathcal{O}(-6), Q) = H^1(Q(6)) = 0$; its splitting is not asserted canonical.

Let $\nu : P = \mathrm{d}P_6 \to W \hookrightarrow X$ be the normalization immersion. There is an exact sequence

$$0 \longrightarrow \mathcal{T} \longrightarrow T_X \longrightarrow \nu_* K_P \longrightarrow 0. \tag{D.10}$$

To verify the quotient, in a chart $W = (z_1 \cdots z_r = 0)$ the map takes a vector field to its normal coefficient on each of the r separate branches. The coefficients lift independently; its kernel is precisely the vector fields tangent to every branch. Thus the quotient is the pushforward of the normal line of ν, not $\mathcal{O}_W(W)$. The form $d\log x_1 \wedge d\log x_2 \wedge dq$ becomes a unit times $dz_1 \wedge dz_2 \wedge dz_3$ in every unimodular chart and is invariant under every cusp deck transformation. It trivializes $\nu^* K_X$. Adjunction for the immersion gives its normal line K_P, proving (D.10). Since P is rational, its K_P cohomology is $(0, 0, 1)$ by Serre duality. Finite pushforward preserves that cohomology.

It remains to identify the extension at p_0 in degree two. We give a fibre-sensitive calculation. The kernel of

$$\Omega^1_X|_W \longrightarrow \nu_*\Omega^1_P$$

is $\nu_*\mathcal{O}_P \otimes N_W^*$. At $xy = 0$ its sections are $a(y, z)y\,dx + b(x, z)x\,dy$; at $xyz = 0$ they are $a(y, z)yz\,dx + b(x, z)xz\,dy + c(x, y)xy\,dz$. These identifications glue by the common conormal dq. Since N_W and $K_X|_W$ are trivial and $H^0(P, \Omega^1_P) = 0$, the Serre-dual space of $H^2(W, T_X|_W)$ is one-dimensional. Top-degree base change identifies this group with the residue fibre of $(R^2\bar{f}_*T_X)_{p_0}$. Its generic rank is one by (D.9). A finite module over the discrete valuation ring $\mathbb{C}\{q\}$ of rank one and residue dimension one is free of rank one. Thus the extension by $\mathbb{C}_{p_0}$ increases the degree of its line part by p_0, without adding torsion. We obtain the complete root direct images

$$R^0\bar{f}_*T_X = \mathcal{O} \oplus \mathcal{O}(-1), \quad R^1\bar{f}_*T_X = \mathcal{O} \oplus \mathcal{O}(-1) \oplus \mathcal{O}(2),$$
$$R^2\bar{f}_*T_X = \mathcal{O}(6) \oplus Q. \tag{D.11}$$

On the order-three chart, an invariant coefficient of the generator of Q must have exponent 2 modulo 3, whereas Q only retains exponents $0, 1$. Therefore $\rho_* Q = 0$. Applying the exact ρ_* to (D.11) and using $d(0) = 0, d(-1) = d(2) = -1, d(6) = 0$ proves (D.4). Finally Leray over a curve has only columns zero and one. The usual cohomology of these displayed line bundles gives the four asserted dimensions. This is a direct-image computation, not an inference from the tangent Euler characteristic. □

Statement D2.1 [Proved]. The entire first tangent cohomology.

The Leray class $\xi \otimes [\gamma]$ generates $H^1(X, T_X)$. The Kodaira–Spencer map of $c \mapsto X_c$ is an isomorphism $T_c\mathcal{P} \to H^1(X_c, T_{X_c})$ at every admissible c.

Proof. All line bundles other than the trivial summands in $R^q\bar{f}_*\mathcal{V} = E \otimes R^q\bar{f}_*\mathcal{O}_X$ are acyclic: they have weights $-1, -2$. Hence $H^1(X, \mathcal{V}) = \mathbb{C}(\xi \otimes \gamma)$ and its other positive-degree groups vanish. The three direct images of $\bar{f}^*\mathcal{O}(-5)$ have weights -5, $(-5, -6)$, and -6, all acyclic by D1.1. Consequently (D.3) induces an isomorphism in every cohomology degree. Sequence (D.10), whose quotient has no cohomology in degrees zero and one, then gives the claimed isomorphism $H^1(\mathcal{V}) \to H^1(T_X)$.

We use the Cech convention in which the derivative of a transition arrow is composed with the inverse of the central arrow. A lattice transition is translation by $\Pi_c\lambda$, and its derivative in c is $\gamma(\lambda)\xi$. The linear deck matrices are independent of c. On the finite charts the affine translations contribute $\gamma(v_j)\xi/m_j$; their m_jth iterates give $\gamma(v_j)\xi$, precisely the lattice cocycle. On the overlap with the smooth part the derivative of the logarithmic translating section is $\gamma(v_j)(\log s_j)\xi/(2\pi i)$. Its Cech coboundary accounts for the difference of these representatives. On the cusp the deck multiplier differentiates to $\gamma(\lambda)\xi$ in logarithmic coordinates, and the exponential overlap identifies it with the same cocycle. Thus the global Kodaira–Spencer class is $[\xi \otimes \gamma]$, with a minus sign if the tangent-lift convention $[\bar{\partial}h]$ is used instead. The sign is recorded; nonvanishing does not depend on it.

Nonvanishing can also be checked on a regular torus. In the quotient description of $H^1(\mathcal{O})$ used above, $[\gamma] \neq 0$; the first coefficient of (D.5) is zero, so $\xi\gamma$ cannot be a scalar multiple of the base Kodaira–Spencer class. Its image in $H^1(T_X|_F)$ therefore remains nonzero. Since $H^1(X, T_X)$ is one-dimensional by D1.3, this proves the asserted isomorphism. □

Statement D3.1 [Proved]. The obstruction map and the Kuranishi germ.

The primary Kuranishi bracket is zero, the Kuranishi germ of X_c is a smooth one-dimensional germ, and the c-family is locally semiuniversal. The equality $h^2(T_X) = h^1(T_X) = 1$ does not identify the nonzero obstruction space with a nonzero obstruction map.

Proof. Let α be the generator in D2.1. The constructed holomorphic family gives an actual convergent deformation tangent to α. Differentiating its integrability equation to second order gives $\bar{\partial}\alpha_2 = -[\alpha, \alpha]/2$, so the cohomology class of $[\alpha, \alpha]$ is zero. Bilinearity proves that the bracket $H^1(T_X) \times H^1(T_X) \to H^2(T_X)$ vanishes on every pair. This argument alone would not exclude higher obstructions, so we make the analytic argument as well. A Kuranishi germ is a closed analytic subgerm of its tangent space $H^1(T_X) = \mathbb{C}$. Its classifying map from the actual parameter disc has nonzero derivative by D2.1. Composing with this embedding gives a holomorphic map of discs with nonzero derivative; the inverse function theorem makes its image contain a neighbourhood of zero in $\mathbb{C}$. Every equation of the Kuranishi germ vanishes on that neighbourhood and is therefore zero. The germ, including its analytic structure, is the smooth line. The same inverse-function argument makes the classifying map an isomorphism of germs. Pulling back the semiuniversal deformation therefore realizes every sufficiently small deformation in the c-family. This is semiuniversality; the positive-dimensional automorphism group is not being discarded or interpreted as a fine-moduli assertion. □

4.1 All integral twists and the generic nonintegral twist

Statement D4.1 [Proved]. Integral Picard twists, including their torsion.

For $M_n = \mathcal{O}_X(n(S_1 - S_2))$, define $u(n) = \max\{d(n) + 1, 0\}$, $v(n) = \max\{-d(n) - 1, 0\}$ and $\epsilon_n = 0$ if $3 \mid n$, $\epsilon_n = 1$ otherwise. Then

$$\begin{aligned} f_*(T_X \otimes M_n) &= \mathcal{O}(d(n)) \oplus \mathcal{O}(d(n-1)), \\ R^1 f_*(T_X \otimes M_n) &= \mathcal{O}(d(n)) \oplus \mathcal{O}(d(n-1)) \oplus \mathcal{O}(d(n+2)), \\ R^2 f_*(T_X \otimes M_n) &= \mathcal{O}(d(n+6)) \oplus \mathbb{C}_{p_1}^{\epsilon_n}. \end{aligned} \tag{D.12}$$

In particular its cusp stalk is free of rank one. The four global dimensions, for every integer n, are

$$\begin{aligned} h^0 &= u(n) + u(n-1), \\ h^1 &= v(n) + v(n-1) + u(n) + u(n-1) + u(n+2), \\ h^2 &= v(n) + v(n-1) + v(n+2) + u(n+6) + \epsilon_n, \\ h^3 &= v(n+6). \end{aligned} \tag{D.13}$$

Proof. The defining roots give $\bar{f}^* R = \mathcal{O}_X(S_1 - S_2)$, so the projection formula applies to (D.11), with its actual coherent torsion summand. Tensoring (D.8) changes its generator character from -1 to $n-1$. There is exactly one invariant coefficient among exponents $0, 1$ if $n-1$ is 0 or 1 modulo three, and none if it is 2. Multiplication by the coarse parameter $t = s^3$ is zero on this module. Its invariant direct image is therefore exactly the displayed skyscraper, of length ϵ_n, not a higher-order thickening. The two other special places contribute no torsion. Formula (D.1) proves (D.12), and the two-column Leray sequence proves (D.13). The code evaluates these formulas for $-240 \le n \le 240$ using two independent integer cohomology counts; their alternating sums are all one. That last check is supplementary to the sheaf proof. □

Statement D4.2 [Proved]. The cusp length and the generic twist.

Use the coordinate $M_z = \exp(z\mathfrak{g})$ of C3.2. For every $z \notin \mathbb{Z}$ the cusp contribution is exactly

$$(R^0 f_*(T_X \otimes M_z))_{p_0} = 0, \quad (R^1 f_*(T_X \otimes M_z))_{p_0} = 0, \quad (R^2 f_*(T_X \otimes M_z))_{p_0} = \mathbb{C}\{q\}/(q). \tag{D.14}$$

Outside the countable exceptional set explicitly defined in D4.3, these are the only nonzero direct-image stalks. Hence the generic twist has $(h^0, h^1, h^2, h^3) = (0, 0, 1, 0)$, and the integer r in Lead 5 equals one there.

Proof. The map $H^1(W, \mathbb{C}) \to H^1(W, \mathcal{O}_W)$ is an isomorphism by the normalization complex in D1.2. The restriction of M_z has trivial pullback to P and descent character $(\exp(2\pi i z), 1)$ on the two generators of the dual torus. The normalization complex with these coefficients is the cellular cochain complex of that rank-one local system. It may be computed on the product cell structure as

$$\mathbb{C} \xrightarrow{(a-1,0)^t} \mathbb{C}^2 \xrightarrow{(0,a-1)} \mathbb{C}, \qquad a = \exp(2\pi i z).$$

For $z \notin \mathbb{Z}$ this is acyclic. All normalization components remain rational with trivial pulled-back twist, so this calculates the actual coherent cohomology $H^q(W, M_z|_W)$, not just a topological analogy. Analytic semicontinuity and base change make $R^q \bar{f}_* M_z$ zero on a neighbourhood of p_0 for all q. The projection formula and (D.3) imply the same vanishing for $\mathcal{T} \otimes M_z$. Tensoring (D.10) remains exact, and $\nu^* M_z = \mathcal{O}_P$ because P is rational and the topological first Chern class of M_z is zero. Thus the degree-two direct image is exactly the reduced

skyscraper $H^2(P, K_P) \otimes \mathbb{C}_{p_0}$. This proves (D.14), including its annihilator and not only its residue dimension.

If the twist is nontrivial on every smooth fibre and on both finite root-cover central tori, all cohomology of its fibre restriction vanishes at those places. Indeed a nontrivial topologically trivial line bundle on a complex torus has zero cohomology: in a unitary flat presentation its Dolbeault Laplacian has no zero Fourier mode. Tensoring the tangent extension by that line preserves acyclicity. Base change on the root charts and exact invariants then give zero direct images away from p_0. D4.3 proves that the excluded twists form a countable set. Leray now gives the asserted four dimensions. □

Statement D4.3 [Proved]. The exceptional nonintegral twists cannot be omitted.

On a regular marked fibre, M_z is trivial if and only if there are integers n, p, q, r such that

$$q\tau(t) + r\mu(t) = p, \qquad z = n - 6q\mu(t) - r\beta_c(t). \tag{D.15}$$

For $z \notin \mathbb{Z}$ these resonances occur in a countable set of parameter values z; every resonant z is nonreal. On the finite root covers the exceptional lattices are

$$\Lambda_1 = \mathbb{Z} + (3\beta_c(z_1) + 4 - 2\rho)\mathbb{Z}, \qquad \Lambda_2 = \mathbb{Z} + (2\beta_c(z_2) + 3 - 3i)\mathbb{Z}. \tag{D.16}$$

Take the union of $\mathbb{Z}$, these two lattices, and all values from (D.15) over regular fibres; this is the countable exceptional set used in D4.2. The set is larger than $\mathbb{Z}$: the regular value $t = 32/5$ supplies nonintegral twists with $h^1(X, T_X \otimes M_z) \geq 2$.

Proof. The exponential sequence on a torus identifies trivial line bundles with integral constant cocycles modulo the two holomorphic period rows. Write $z\gamma = (n, p, q, r) + a(6\mu, \tau, 1, 0) + b(\beta_c, \mu, 0, 1)$. The third and fourth coordinates give $a = -q, b = -r$; the first two give exactly (D.15). If $r = 0$, the positivity of $\operatorname{Im}\tau$ forces $q = p = 0$, leaving $z = n$. Otherwise the imaginary parts of (D.15) give

$$\operatorname{Im}\mu = -\frac{q}{r}\operatorname{Im}\tau, \qquad \operatorname{Im} z = -rD \neq 0. \tag{D.17}$$

For fixed (p, q, r) with $(q, r) \neq (0, 0)$, the first equation of (D.15) has discrete zeros, since $1, \tau, \mu$ are linearly independent. A discrete subset of this second-countable Riemann surface is countable. There are countably many integer tuples and markings, so the union of the resulting z is countable.

At z_1, insert $\tau = \rho$, $\mu = (2 - \rho)/3$ in the first equation. Its solutions have $(q, r, p) = (k, 3k, 2k)$. At z_2 they have $(q, r, p) = (k, 2k, k)$. The second equation gives (D.16). These are discrete lattices because their nonintegral generators have imaginary parts $3D$ and $2D$.

For the explicit regular resonance, use the section $P = (0, \sqrt{2}t^2(t-1))$ of the stated elliptic surface. The chord-and-tangent law gives

$$x(2P) = 9t^2/8, \qquad y(2P) = t^2(32 - 5t)/(16\sqrt{2}).$$

At $t = 32/5$, the discriminant $1728t^8(t-1)^3$ is nonzero, P is not two-torsion, and $2P$ is a nonzero two-torsion point. Thus P has exact order four. The period identification in Part A supplies a nonzero relation $q\tau + 4\mu = p$ with integers p, q. Equation (D.15), with $r = 4$, defines an actual twist for which the regular torus restriction is trivial; (D.17) shows that it is nonintegral. The period derivative there has nonzero determinant, so the tangent restriction has $h^0(T_X|_F) = 2$. The degree-zero base-change exact sequence gives

$$H^0(F, (T_X \otimes M_z)|_F) \simeq (R^1 f_*(T_X \otimes M_z))_t[t],$$

because $f_*(T_X \otimes M_z) = 0$ for a twist nontrivial on the general fibre. Its right side has dimension two. The degree-one direct image is a finite-support sheaf, so its global sections have dimension at least two, proving the bound on h^1. The exact elliptic calculation is checked by both duplication and substitution in `d_exact.py`; its output records the nonzero rational discriminant. The choice of integers p, q depends on the marked elliptic logarithm, and no unverified numerical marking is substituted. □

Statement D4.4 [Proved]. Local complexes and every noncusp torsion length.

Let $z \notin \mathbb{Z}$. At a regular resonance t_* in (D.15), with $A = \mathbb{C}\{t - t_*\}$, the two stalks are

$$(R^1 f_*(T_X \otimes M_z))_{t_*} = (A/(t - t_*))^2, \qquad (R^2 f_*(T_X \otimes M_z))_{t_*} = A/(t - t_*)^2. \tag{D.18}$$

At p_1, write $z = n + k(3\beta_1 + 4 - 2\rho)$, $k \neq 0$, when this is possible. Both coarse stalks are killed by t and have length $1 + \epsilon_n$, with ϵ_n as in D4.1. At p_2, write $z = n + k(2\beta_2 + 3 - 3i)$, $k \neq 0$, when possible. Both coarse stalks are killed by $t - 1$ and have length one for $n \equiv 1, 2 \pmod 4$ and length zero for $n \equiv 0, 3 \pmod 4$. If the relevant lattice representation does not exist, both stalks at that finite fibre are zero.

Proof. We first give the comparison that turns the local calculation into a computation of the sheaves. On a simply connected disc of smooth torus fibres, including a finite root cover, retain the real marking and the holomorphic period matrix. Translation by $\Pi(s)v$, for fixed $v \in \mathbb{R}^4$, is holomorphic on the total family. After subtracting the integer cocycle of a resonance, the line bundle has constant lattice multipliers. Multiplying the lifted real translation by its constant factor of automorphy makes the lattice act as the identity and gives a genuine compact real torus action on the line bundle. The same action acts on the tangent bundle by differentiation. It commutes with the full Dolbeault differential.

Averaging over this compact torus projects the relative Dolbeault complex onto its invariant part. Choose the lift so that the central trivial line has trivial character. On every nonzero Fourier character, the relative Dolbeault symbol is exterior multiplication by a nonzero covector; contraction by a dual covector is a contracting homotopy. After shrinking the disc, these nonzero covectors have a uniform positive lower bound for bounded Fourier frequency and grow linearly for unbounded frequency, because the real period matrices and their inverses stay bounded. The homotopies therefore preserve smooth Fourier series, with all derivatives, and are uniform on compact subsets of the disc. The tangent extension is triangular with respect to its vertical and horizontal parts; its nilpotent off-diagonal term is treated by the finite homological perturbation series, so every nonzero Fourier complex is still contractible. Thus averaging is a quasi-isomorphism, not merely a fibrewise dimension comparison.

The chain-level comparison can be written without an unspecified holomorphic gauge. Write the normalized period matrix as $[Z(s), I]$ and use the real coordinates $\zeta = Z(s)x + y$, with $x, y \in \mathbb{R}^2/\mathbb{Z}^2$. Set

$$G_j = (dx_j)^{0,1}, \qquad e_i = \partial_{\zeta_i}, \qquad h = \partial_s + \sum_i (Z'(s)x)_i e_i.$$

These are invariant under the real translations. Since each dx_j is closed on the total real product, $\bar\partial G_j = 0$. The forms G_j frame the relative invariant $(0,1)$ forms because $\operatorname{Im} Z$ is invertible, and

$$\bar\partial h = \sum_{i,j} Z'_{ij}(s) e_i G_j, \qquad (dy)^{0,1} = -Z(s)(dx)^{0,1}.$$

For the smooth line frame defined by its flat character, the last identity gives $\bar\partial \ell = 2\pi i(a_1 G_1 + a_2 G_2)\ell$, with a_1, a_2 in (D.20); there is no additional base term. All displayed coefficients are

holomorphic in s. The full invariant complex is thus the ordinary base Dolbeault resolution of the following finite complex of holomorphic bundles:

$$C^j = (V \oplus L) \otimes \bigwedge^j G, \quad 0 \le j \le 2, \qquad d(v \otimes \eta) = v \otimes a \wedge \eta,$$
$$d(h \otimes \eta) = h \otimes a \wedge \eta + \kappa \wedge \eta. \tag{D.19}$$

Here V is the rank-two vertical bundle, L the horizontal line, $G = R^1\mathcal{O}$ on the root chart, and $\kappa \in V \otimes G \otimes L^*$ is the actual tangent-extension cocycle. The matrix is obtained by differentiating the period translations. In the raw period and constant-cocycle frames it is

$$\kappa = \begin{pmatrix} 6\mu' & \tau' \\ \beta_c' & \mu' \end{pmatrix}, \qquad a = (z - n + 6q\mu + r\beta_c, -p + q\tau + r\mu). \tag{D.20}$$

The conventional factor $2\pi i$ in the Dolbeault representative of the exponential class multiplies both components of a by the same nonzero constant; absorbing it in the frame has no effect on any module below. The displayed formula also follows directly by subtracting the holomorphic period rows from the constant lattice cocycle. This identifies the chain map and its tangent extension; no unspecified map between fibre cohomology groups is being used. For the nonzero Fourier characters, use the filtration by the number of base $(0,1)$ factors: the relative contractions make the associated graded complexes acyclic, and the two-step filtration then proves acyclicity of the full complexes. Thus the contraction need not commute with the base Dolbeault operator. The finite cyclic group normalizes the real translation group, so averaging and this comparison are equivariant; exact invariants compute the actual multiple-fibre stalks.

At a regular resonance, $a' = \kappa^t(q,r)^t$ and $(q,r) \neq (0,0)$. The determinant of κ is nonzero there by Part A. Therefore a has a simple zero. Choose its nonzero leading direction as the second frame of G and use invertibility of κ to choose the two vertical frames. Holomorphic invertible changes reduce (D.19) to

$$a = (0, s), \qquad \kappa = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \tag{D.21}$$

The second cohomology is generated by $h\gamma\delta$ and is killed by s^2; the relations eliminate the two vertical generators. The first cohomology is generated by $e_1\delta, e_2\delta$, each killed by s. These are equalities of modules over $\mathbb{C}\{s\}$ and prove (D.18).

There is an additional joint coefficient to check at the order-three point. Put $t = s^3$, $\nu = \mu - (2-\tau)/3$ and $b = \beta_c + 4\mu + (2/3)\tau - b(0)$. The order-three transformation laws give $\nu = \nu_2 s^2 + O(s^3)$ and $b = b_3 s^3 + O(s^4)$. The elliptic calculation in A2.3 gives, with elliptic scale $f(s) = Fs^2 + O(s^3)$,

$$\nu_2^2 = 9F^2/512.$$

For clarity we evaluate the other coefficients with the original Weierstrass normalization. Since $x = f^2\wp$, $y = f^3\wp'/2$, one has $f^4 g_2 = 12t^3(t-1)$ and $f^6 g_3 = -8t^4(t-1)^2$. The identities $g_2 = 4\pi^4 E_4/3$, $g_3 = 8\pi^6 E_6/27$ and $E_4'(\rho) = -2\pi i E_6(\rho)/3$ give

$$E_6(\rho) = -\frac{27}{\pi^6 F^6}, \qquad \tau(s) = \rho + \tau_1 s + O(s^2), \quad \tau_1 = \frac{i\pi F^2}{2}.$$

The exact period determinant identity in Part A gives

$$K(s) = 6\mu'^2 - \tau'\beta_c' = K_2 s^2 + O(s^3), \quad K_2 = \frac{3F^2}{4}.$$

Equivalently $K = 6\nu'^2 - \tau' b'$, so $K_2 = 24\nu_2^2 - 3\tau_1 b_3$ and $b_3 = 7i/(32\pi)$.

For a nonintegral resonance the relation is $(q,r,p) = \pm(k,3k,2k)$, $k \neq 0$; the sign only changes the following nonzero scalar. After a constant change of the two cocycle frames, its Picard section is $3k(\nu, b)$, and the transpose period derivative becomes

$$\begin{pmatrix} \tau' & \nu' \\ 6\nu' & b' \end{pmatrix}.$$

Eliminating the unit τ' leaves the joint transverse component $3k(b - 6\nu\nu'/\tau')$. Its coefficient of s^3 is

$$-\frac{k(K_2 + 12\nu_2^2)}{\tau_1} = \frac{123ik}{64\pi} \neq 0. \tag{D.22}$$

Thus neither the order of the Picard section nor the determinant order alone is used to select a Smith form. The joint coefficient has been evaluated exactly.

Choose the second cocycle frame to be the Picard section divided by s^2. Equivariance and (D.22) give a matrix with a unit in its lower-right entry, first-order off-diagonal terms with nonzero coefficients, and a second-order upper-left term. Subtracting a holomorphic multiple of the second vertical row removes the last term; the nonzero determinant coefficient and (D.22) keep both off-diagonal coefficients units. Rescaling the equivariant frames therefore reduces (D.19) to

$$a = (0, s^2), \qquad \kappa = \begin{pmatrix} 0 & s \\ s & 1 \end{pmatrix}. \tag{D.23}$$

Its cohomology is

$$H^1 = \mathbb{C}\{s\}/(s^2) \oplus \mathbb{C}\{s\}/(s^3), \qquad H^2 = \mathbb{C}\{s\}/(s^2) \oplus \mathbb{C}\{s\}/(s^3). \tag{D.24}$$

For example in degree one put $u = e_1\delta$, $v = e_2\gamma + sh\delta$; the remaining cycle $e_2\delta$ is eliminated by the boundary $su + e_2\delta + sv$. The two surviving relations are $s^2u = 0, s^3v = 0$. In degree two eliminate $e_2\gamma\delta$; the remaining generators $e_1\gamma\delta, h\gamma\delta$ have annihilators s^2, s^3 respectively.

We specify the character convention to avoid a sign ambiguity. A frame has invariant exponent w when its coefficient satisfies $g(\zeta s) = \zeta^w g(s)$; its pullback-frame weight is the negative of w. For the trivialized twist in (D.16), these exponents are n at p_1 and $3n$ at p_2. This agrees with M_n in D4.1 and follows by evaluating its integer cocycle on v_1, v_2. At p_1 the two generators in H^1 of (D.24) have invariant exponents $n+2, n+2$, and the two generators in H^2 have exponents $n+2, n$. The length-three summand has one invariant coefficient for every n; the length-two summand has one precisely when $n \not\equiv 0 \pmod 3$. The coarse parameter is s^3, so all these invariants are killed by it. This proves the asserted order-three stalks with the full nilpotent layers included.

At p_2 the determinant is a unit. The Picard section has a simple zero, so (D.21) applies on the order-four cover. The invariant exponents of the two degree-one generators are $3n+1, 3n+2$; the cyclic degree-two module of length two has generator exponent $3n+2$. In both degrees its invariant dimension is one exactly for $n \equiv 1, 2 \pmod 4$. The coarse parameter s^4 kills the modules. This proves the stated order-four calculation. The two independent exact checks in `d_koszul.py` are Smith normal forms and ranks of truncated jet complexes; the latter are also restricted to every stabilizer character. Their outputs agree with (D.21)–(D.24) and with every invariant length just computed. □

Statement D4.5 [Proved]. The full twisted cohomology and its jump loci.

For $z \notin \mathbb{Z}$, let $\mathcal{R}_z$ be the set of regular base points satisfying (D.15), counted as points, not as marked integer tuples. It is finite. Put $N_z = |\mathcal{R}_z|$. Define $\ell_1(z) = 0$ off Λ_1, and $\ell_1(z) = 1 + \epsilon_n$ if $z = n + k(3\beta_1 + 4 - 2\rho)$, $k \neq 0$. Define $\ell_2(z) = 1$ if $z = n + k(2\beta_2 + 3 - 3i)$, $k \neq 0$, with $n \equiv 1, 2 \pmod 4$, and $\ell_2(z) = 0$ otherwise. Then, setting $J_z = 2N_z + \ell_1(z) + \ell_2(z)$,

$$\boxed{(h^0, h^1, h^2, h^3)(T_X \otimes M_z) = (0, J_z, 1 + J_z, 0).} \tag{D.25}$$

More precisely, with $\mathfrak{m}_t$ the maximal ideal at t,

$$R^1 f_*(T_X \otimes M_z) = \bigoplus_{t \in \mathcal{R}_z} \mathbb{C}_t^2 \oplus \mathbb{C}_{p_1}^{\ell_1(z)} \oplus \mathbb{C}_{p_2}^{\ell_2(z)},$$

$$R^2 f_*(T_X \otimes M_z) = \mathbb{C}_{p_0} \oplus \bigoplus_{t \in \mathcal{R}_z} \mathcal{O}_{B,t}/\mathfrak{m}_t^2 \oplus \mathbb{C}_{p_1}^{\ell_1(z)} \oplus \mathbb{C}_{p_2}^{\ell_2(z)}. \tag{D.26}$$

Together with D4.1 these formulas cover every element of $\mathrm{Pic}(X) = \mathbb{C}$. The actual cohomology jump loci are locally discrete subsets of this complex line.

Proof. The support assertions and all stalks were proved in D4.2–D4.4. The direct image in degree zero is zero for a nonintegral twist: it has generic rank zero and is torsion-free, since multiplication by a nonzero base function is injective on the bundle on the smooth integral total space. The support of either higher coherent direct image is a proper closed analytic subset of the compact curve, and hence is finite. This proves finiteness of $\mathcal{R}_z$ as well as the absence of an accumulation at the special values. If two integer tuples at a marked fibre represented the same trivialized line, their difference would be an integral real class in $H^{1,0}$, which is zero. Thus there is no multiplicity from duplicate tuples. Formula (D.26) follows by the decomposition of a finite-support sheaf into its local modules. Its global sections have dimension equal to its length and its H^1 is zero, so Leray gives (D.25).

To clarify the word generic, the global closed form G_γ in C3.8 defines a holomorphic line bundle on $X \times \mathbb{C}$ by the integrable operators $\bar\partial - 2\pi i z G_\gamma$. Proper coherent base change for the projection to $\mathbb{C}$ makes each cohomology-dimension threshold an analytic subset. A real noninteger z has no resonance by (D.17) and (D.16), so the generic values are attained. All possible exceptional values were shown to form a countable set. Thus each actual threshold exceeding a generic dimension is a proper analytic subset of a complex curve and is locally discrete. Some indexed resonances, namely the two invisible order-four congruences, produce no jump; (D.25) distinguishes them.

This is an explicit countable description of the jump loci in the Picard coordinate, rather than a generic assertion. It can also be indexed algebraically on the elliptic surface. If a regular fibre has torsion section P of exact order r_t, choose its marked relation $q_t\tau + r_t\mu = p_t$. Its full resonance lattice is

$$\mathbb{Z} + \big(6q_t\mu(t) + r_t\beta_c(t)\big)\mathbb{Z}.$$

Every regular resonance arises this way. The underlying values t are exactly the non-special roots of the division polynomials $\psi_N(P)$, $N \geq 2$. The nonintegral lattice points contribute two to J_z for each distinct such fibre. The finite-point lattices have the congruence weights already stated. The script `d_division_polynomials.py` evaluates the polynomials exactly for $2 \leq N \leq 12$; for example the first nonconstant regular factors are $5t - 32$, $89t^2 + 4736t - 4096$ and $11t^2 - 6016t + 8192$. These computations specify an executable enumeration of the resonance loci without inventing decimal values for the unspecified parameter c.

As an independent index check, $c_1(X) = c_2(X) = c_1(M) = 0$ in integral cohomology, so the degree-six Riemann–Roch integrand for the actual twisted tangent bundle is $c_3/2$. Its integral is one. Both (D.13) and (D.25) have alternating sum one. The index is used here to verify the completed sheaf calculation, not to select its lengths. □

Provenance of the tangent argument. The search located Du, Guo, Kusakabe, and Xie, arXiv:2610.00250v1 Du et al. (2026a), whose Section 6 asserted the same untwisted tangent dimensions and smooth one-dimensional deformation germ. The current v2 Du et al. (2026b) omits that section and those assertions. We do not use the superseded theorem as an established input. The root-base strategy above was prompted by that section; we have supplied its local comparisons and the additional coarse direct-image and integral-twist calculation. We claim no priority. This provenance paragraph is distinct from the proofs and from the verified status of the current paper.

5 E. Small deformations and the comb construction

Statement E1.1 [Proved]. Stability of the pencil for the standard family.

Every sufficiently small deformation of the standard X_c carries a holomorphic map to $\mathbb{P}^1$ with general fibre a complex two-torus. Its complete anticanonical system $|-2K|$ is basepoint-free of dimension one and recovers that map, and its algebraic dimension is one.

Proof. D3.1 identifies the semiuniversal germ with the actual c-parameter germ. Thus every sufficiently small deformation is isomorphic to a member $X_{c'}$ with $\operatorname{Im} c' < -M$. The bundle identity $K_X = \bar{f}^*\mathcal{O}_{\mathcal{B}}(-6) = \mathcal{O}_X(-2S_2)$ in D1.2 holds relatively in this family: all root divisors and every determinant transition used in its proof vary holomorphically, and their characters are constant. Consequently $-2K = \bar{f}^*\mathcal{O}(12) = f^*\mathcal{O}_B(1)$ on every member, with a relative line-bundle isomorphism. Connected fibres imply $f_*\mathcal{O}_X = \mathcal{O}_B$, so the two base sections pull back to the entire two-dimensional section space. They have no common zero. The resulting map is the constructed f followed by a projective coordinate change. Its general fibre is a torus by the same local quotient construction.

We spell out the algebraic-dimension reduction used here. On a manifold with $H^2(X,\mathbb{R}) = 0$, a divisor restricts to a topologically trivial line bundle on any regular torus fibre. A nonzero effective divisor on a compact torus has positive degree against a Kahler form, whereas a topologically trivial class has degree zero. Hence an effective divisor on X cannot meet a regular fibre in a proper nonempty divisor: its restriction is either the whole fibre or empty. The zero and pole divisors of a meromorphic function are therefore vertical. On a regular fibre where neither contains the fibre, the restricted function has no pole and is constant. The function hence descends meromorphically to the base on a dense open set, and the proper connected-fibre map extends the descent across the finitely many special values. Thus the meromorphic field is $\mathbb{C}(t)$ and $a(X) = 1$. This argument applies to each small member and does not appeal to an invalid vanishing theorem.

The conclusion is stronger than what semicontinuity alone gives. For the original member, projection formula gives $h^\bullet(f^*\mathcal{O}(1)) = (2,3,1,0)$; the nonzero degree-one group means that a generic line-bundle extension argument would not establish stability of its two sections. Here stability follows from the computed semiuniversal family and its relative root divisors. □

Statement E2.1 [Known]. The structure sheaves of the comb members.

Let X_m be any sufficiently small smooth period section obtained by the m-tooth construction of Viaclovsky (2026a, Theorem 1.1), with $m \geq 12$, fixed limiting data at the three distinguished values, and the standard logarithmic-transformation data. Write $f_m : X_m \to B$. Then

$$R^1 f_{m*}\mathcal{O}_{X_m} = \mathcal{O}(a_m) \oplus \mathcal{O}(b_m), \quad a_m, b_m \geq 0, \quad a_m + b_m = 2m-1,$$
$$R^2 f_{m*}\mathcal{O}_{X_m} = \mathcal{O}(2m-1), \qquad (h^{0,0}, h^{0,1}, h^{0,2}, h^{0,3}) = (1, 2m+1, 2m, 0). \tag{E.1}$$

The individual summands are not specified by m in the construction.

Proof. The paper's equation (7.5) proves that attaching and smoothing one twistor conic increases the determinant degree of its quotient bundle by two. Its Proposition 9.1 identifies the initial degree as -1, so the smoothed quotient has degree $2m-1$. The geometric gluing comparison in its Section 6.3 identifies that quotient with $R^1 f_{m*}\mathcal{O}_{X_m}$; the fixed limiting periods preserve the same cyclic representations at the finite points and the same normalization complex at the cusp. Thus its Theorem 2.3 applies to these actual full scheme-theoretic fibres and gives the cup-product isomorphism $\det R^1 f_{m*}\mathcal{O} = R^2 f_{m*}\mathcal{O}$. Its Proposition 3.2, whose hypotheses are connected fibres, torus general fibre and the topology of S^6, gives the two possible splitting alternatives: both degrees nonnegative, or $(-1,0)$. The positive determinant degree excludes the latter. We have now derived the stated sheaves from the source comparisons in precisely the form needed here. Leray on B gives $h^{0,1} = a_m + b_m + 2$, $h^{0,2} = a_m + b_m + 1$ and $h^{0,3} = 0$, proving (E.1). The theorem is used for the smoothed section, not for the singular comb, which is not a small deformation of X_0. □

Statement E2.2 [Proved]. Canonical bundle, exact tangent cohomology and Hodge growth on the combs.

For the same X_m, the root lift $\bar{f}_m : X_m \to \mathcal{B}$ has vertical bundle E_m and $G_m = R^1 \bar{f}_{m*}\mathcal{O}_{X_m}$ satisfying

$$0 \longrightarrow \mathcal{O} \longrightarrow E_m \longrightarrow \mathcal{O}(24m-1) \longrightarrow 0, \qquad 0 \longrightarrow \mathcal{O} \longrightarrow G_m \longrightarrow \mathcal{O}(24m-1) \longrightarrow 0. \tag{E.2}$$

The canonical bundle and all positive-degree holomorphic forms obey

$$K_{X_m} = \bar{f}_m^* \mathcal{O}(-24m-6), \qquad h^{1,0} = h^{2,0} = h^{3,0} = 0. \tag{E.3}$$

Moreover

$$\boxed{h^\bullet(T_{X_m}) = (2m+1, 8m+1, 6m+1, 0).} \tag{E.4}$$

More precisely, writing $\mathcal{L} = \mathcal{O}_{\mathcal{B}}(-5)$,

$$\begin{aligned} f_{m*}T_{X_m} &= \rho_* E_m, \\ R^1 f_{m*}T_{X_m} &= \operatorname{coker}\{\rho_*\mathcal{L} \xrightarrow{\delta_0} \rho_*(E_m \otimes G_m)\}, \\ R^2 f_{m*}T_{X_m} &\simeq \mathcal{O}_B(2m) \oplus Q_m, \qquad \operatorname{length} Q_m = 4m, \end{aligned} \tag{E.4a}$$

where $Q_m = \operatorname{coker}\{\rho_*(\mathcal{L}\otimes G_m) \xrightarrow{\delta_1} \rho_*(E_m \otimes \det G_m)\}$ has no support at the cusp. The splitting in the last line need not be canonical. There is a rank-three vector bundle K_m on B for which

$$\boxed{h^{1,1}(X_m) = 4m + 1 + h^0(K_m), \qquad h^{1,2}(X_m) = 4m + h^0(K_m).} \tag{E.5}$$

In particular both remaining middle Hodge numbers grow at least linearly with m.

Proof. The construction retains the integral local system and the limiting period planes. The invariant real homology class $\widehat{\delta}$ therefore gives a vertical translation field on every regular torus. It is nowhere zero there, since the real period map is an isomorphism. At a finite root point the same reasoning applies on the smooth cover, and at the cusp its nonzero angular class gives a nonzero logarithmic vertical vector. The local vertical transitions, which depend only on the base exactly as in D1.2, define E_m and give its trivial subbundle. The invariant real cohomology class γ gives a nowhere-zero section of G_m, by the same smooth-fibre and cusp comparison in D1.2.

The determinant of G_m has the same two stabilizer characters as that of G_0: invariant exponents 2 at p_1 and 1 at p_2. Its coarse determinant has degree $2m-1$ by E2.1. The explicit

descent formula in D1.1 therefore identifies $\det G_m = \mathcal{O}(12(2m-1)+8+3) = \mathcal{O}(24m-1)$. The rank-four flat bundle on the root base has trivial determinant: the integral monodromies have determinant one, the finite-root charts have no residual ramification, and the logarithmic cusp monodromy has nilpotent residue of trace zero. Its flat volume section thus extends without a zero or pole. The exact Hodge-plane sequence $0 \to E_m^* \to \mathcal{H} \to G_m \to 0$ gives $\det E_m = \det G_m$. Taking quotients by the two nonvanishing sections proves (E.2). The determinant of the log tangent sequence $0 \to \bar{f}_m^* E_m \to T_{X_m}(-\log W) \to \bar{f}_m^* \mathcal{O}(-5) \to 0$ then gives (E.3)'s canonical formula.

The logarithmic Kodaira–Spencer map is invertible near the cusp. This does not follow merely from fixing a limiting plane: Viaclovsky's Proposition 6.1 gives the actual local period expression $(\log q) I/(2\pi i) + H_m(q)$ with H_m holomorphic. Its logarithmic derivative has invertible leading matrix $I/(2\pi i)$, or $B_0/(2\pi i)$ after the retained marking change. Thus the map is generically invertible. The direct image of the logarithmic cotangent sequence consequently has kernel exactly $\mathcal{O}(5)$. It has no section by D1.1, so $h^{1,0} = 0$. To treat two-forms, take the second exterior power of that sequence. Its possible global sections lie in $H^0(E_m^* \otimes \mathcal{O}(5))$ or $H^0((\det E_m)^{-1})$. The latter is zero, having weight $1 - 24m < 0$. A morphism $E_m \to \mathcal{O}(5)$ vanishes on the subline $\mathcal{O}$, since $H^0(\mathcal{O}(5)) = 0$, and therefore factors through $\mathcal{O}(24m-1)$. The resulting section would have weight $6 - 24m < 0$, so it too vanishes. Absolute two-forms inject into logarithmic ones, which proves $h^{2,0} = 0$. The negative canonical weight proves $h^{3,0} = 0$.

The first connecting map from the horizontal line to $E_m \otimes G_m$ is nonzero near the cusp, so it is injective as a sheaf map. The same normal quotient (D.10) has no degree-zero direct image, giving $f_{m*} T_{X_m} = \rho_* E_m$. Exact invariants applied to (E.2) give, for each of $V = E_m, G_m$,

$$0 \longrightarrow \mathcal{O}_B \longrightarrow \rho_* V \longrightarrow \mathcal{O}_B(2m-1) \longrightarrow 0.$$

These bundles have vanishing H^1 and $h^0 = 2m+1$. They also split into two nonnegative-degree lines: if a splitting had one negative summand, the section of its saturated $\mathcal{O}_B$ subline would have to lie in the positive summand. Every nonzero section of that summand has a zero, contradicting saturation.

The natural map

$$(\rho_* E_m) \otimes (\rho_* G_m) \longrightarrow \rho_*(E_m \otimes G_m)$$

is injective, with length-one cokernel at p_1 and no other cokernel. Indeed both bundles have invariant exponents $(0,2)$ at p_1. The product of their exponent-two generators is s^4, whereas the product representation has invariant generator s; the comparison on that one summand is multiplication by $t = s^3$. All other exponents add without a carry. At p_2 the exponents are $(0,1)$, and no sum reaches four. At the cusp there is no stabilizer. The source is a sum of nonnegative-degree line bundles, so its H^1 is zero. The length-one upper modification consequently has $H^1 = 0$, rank four and degree $2(2m-1) + 2(2m-1) + 1 = 8m-3$; it has $8m+1$ global sections.

In the direct-image sequence of the log tangent extension, the second cup map $\delta_1 : \mathcal{L} \otimes G_m \to E_m \otimes \det G_m$ is generically invertible, by the invertible cusp Kodaira–Spencer matrix. It is therefore injective as a sheaf map. The first cup map δ_0 is injective for the same reason. Consequently $R^1 \bar{f}_{m*} T(-\log W) = \operatorname{coker} \delta_0$. The normal quotient $\nu_* K_P$ in (D.10) has zero direct images in degrees zero and one, so the same formula holds for $R^1 T$. Since $\rho_* \mathcal{L} = \mathcal{O}_B(-1)$ is acyclic, its cokernel has $h^0 = 8m+1$ and $h^1 = 0$. Leray and $H^1(\rho_* E_m) = 0$ now give the exact value $h^1(T) = 8m+1$.

For completeness the degree-two calculation gives (E.4) directly, with the full torsion correction retained. The coarse degrees of $\mathcal{L} \otimes G_m$ and $E_m \otimes \det G_m$ are $2m-2$ and $6m-2$. Their invariant exponents at p_1 are respectively $(1,0)$ and $(2,1)$, and at p_2 both are $(1,2)$. Their root determinant weights are $24m-11$ and $72m-3$; subtracting the character contributions

and dividing by twelve gives exactly these coarse degrees. Thus Q_m has length $4m$. It has no cusp contribution, where the cup map is invertible. On the root base there is an exact sequence

$$0 \longrightarrow \operatorname{coker}\delta_1 \longrightarrow R^2\bar{f}_{m*}T(-\log W) \longrightarrow \mathcal{L}\otimes\det G_m = \mathcal{O}(24m-6) \longrightarrow 0.$$

It splits, since the extension group is the first cohomology of a zero-dimensional sheaf. The cusp normal quotient then raises the degree of the free line by p_0, exactly as in D1.3. The local fibre, its normalization and the logarithmic residue are unchanged; in particular $h^2(T|_W)=1$ still excludes a split torsion summand at the cusp. Hence the new free line is $\mathcal{O}(24m+6)$, which descends to $\mathcal{O}_B(2m)$, proving (E.4a). Leray gives $h^2(T)=(2m+1)+4m=6m+1$ and $h^3(T)=0$. As an independent check, these values satisfy the full tangent index $\chi(T)=1$. The source family of Viaclovsky's Proposition 8.3 also gives

$$8m-8 \le \dim \operatorname{Def}(X_m) \le 8m+1;$$

the lower bound concerns the actual classifying-family dimension, while the upper bound is the tangent dimension now computed.

We give the bundle specifying the remaining Hodge number. Set

$$K_m = \ker\left\{\rho_*(E_m^*\otimes G_m) \xrightarrow{\kappa_1} \rho_*(\mathcal{O}(5)\otimes\det G_m)\right\}. \tag{E.6}$$

The map is cup product with the actual logarithmic Kodaira–Spencer class, in the category of coherent sheaves on B. It is generically onto, so the kernel is a rank-three vector bundle. The preceding connecting map

$$\rho_*E_m^* \xrightarrow{\kappa_0} \rho_*(\mathcal{O}(5)\otimes G_m)$$

is injective. The source and target have degrees $-2m-1$ and $2m-1$. To check these degrees, use their fixed local character exponents $(0,1)$ versus $(2,1)$ at the order-three point and $(0,3)$ versus $(3,0)$ at the order-four point, together with the root determinant degrees. The invariant determinant comparison in D1.1 gives precisely the displayed integers. Its cokernel A_m therefore has length $4m$, irrespective of how its zeros are distributed among regular and special fibres.

Writing $\mathcal{A}_m=\Omega^1_{X_m}(\log W)$, the direct-image sequence gives $0\to A_m\to R^1f_{m*}\mathcal{A}_m\to K_m\to 0$. The residue sequence $0\to\Omega^1_{X_m}\to\mathcal{A}_m\to\nu_*\mathcal{O}_P\to 0$ has a surjective degree-zero residue map at p_0: the logarithmic horizontal form dq/q has residue one on every branch. Its kernel is the one elementary modification $f_{m*}\Omega^1=\mathcal{O}_B(-2)$ of $f_{m*}\mathcal{A}_m=\rho_*\mathcal{O}(5)=\mathcal{O}_B(-1)$. The rationality of P makes the higher direct images of the residue quotient zero. Thus $R^1f_{m*}\Omega^1=R^1f_{m*}\mathcal{A}_m$. Since $H^1(A_m)=0$, its global sections have dimension $4m+h^0(K_m)$, and Leray adds $h^1(\mathcal{O}_B(-2))=1$ in total degree one. This proves the formula for $h^{1,1}$. Finally the sphere Chern numbers give $\chi(\Omega^1)=-1$, and the vanishing just proved, together with Serre duality, gives $h^{1,1}-h^{1,2}=1$. This proves (E.5). □

Statement E2.3 [Open]. The remaining comb splitting and deformation calculations.

The integer m and the stated limiting data do not determine in this paper the splitting of the rank-three bundle K_m in (E.6), its exact number of sections, or the obstruction map of the comb member. These parts of E.2 are unfinished. The four tangent cohomology dimensions themselves are fully determined by (E.4).

The remaining finite data and the computations made. If e_m is the length of $\operatorname{coker}\kappa_1$ in (E.6), then its source has rank four and degree -2, its target is $\mathcal{O}_B(2m)$, and

$$\deg K_m = -2m-2+e_m. \tag{E.7}$$

The degree -2 follows from the two pairs of opposite stabilizer characters of $E_m^* \otimes G_m$; its root determinant is trivial, and each finite root point subtracts one from the coarse determinant. Although (E.4a) determines the tangent dimensions independently of the individual zeros of the Kodaira–Spencer determinant, the support and elementary divisors of Q_m still depend on that period-section map. The comb Kuranishi problem further requires its actual bracket and higher obstruction maps; the dimensions in (E.4) do not determine those maps.

We evaluated the universal exact degree and dimension expressions for $m = 12, 13, 20, 100$ in `d_exact.py` and `d_comb.py`; the first case already gives $h^{1,1} \geq 49$, $h^{1,2} \geq 48$, $h^0(T) = 25$, $h^1(T) = 97$ and $h^2(T) = 73$. The smallest undecided instance is the splitting of (E.6) for one specified smoothed 12-tooth period section, together with that member's obstruction map. The attached construction supplies an existence argument and an open set of smoothing choices, not the transition matrices of a uniquely specified X_{12}. We have not chosen and solved its nonlinear period-section smoothing equations, and have not run two numerical methods on those missing matrices. Formulas (E.5)–(E.7) and the growth bounds are proved, but they are not replacements for that remaining calculation or for its Kuranishi bracket. □

Statement E2.4 [Refuted]. The universal two-vector-field bound.

The assertion that every complex structure on S^6 has $h^0(T) \leq 2$, printed in Huckleberry et al. (2000, Corollary 1.3; verified preprint p. 2) and repeated in Claude and Alpöge (2026, Remarks 1.2 and 9.26), is false for the comb members.

Proof. E2.2 gives $h^0(T_{X_m}) = 2m + 1 \geq 25$ for $m \geq 12$. The source proof expresses a third field as a meromorphic linear combination of two fields and invokes the Campana–Demailly–Peternell claim that all meromorphic functions on a complex six-sphere are constant. That implication is unavailable here: f_m supplies nonconstant meromorphic functions. The independently computed $2m + 1$ sections above give the counterexample, so this is a refutation of the bound as well as a location of the failed step. It does not refute a statement restricted to algebraic dimension zero. □

6 Computations, source verification, and scope of the remaining gaps

All new arguments and code in this dossier are AI-produced by GPT-6 Astra (OpenAI, ChatGPT Pro mode). We claim no priority. The supplied original manuscript, its transcription, the workbench, course and reader notes were treated as claims to check. The source manifest contains forty file hashes, all of which were verified before extraction. Where a transcribed formula mattered, we also inspected the original PDF; the source-image checks include pages 5, 62–63 and 81. The local correction in A4.1 concerns the original text itself, not merely OCR.

The exact computations use integer and rational arithmetic in Python and SymPy. The program `b_dictionary.py` verifies the integral monodromy conjugations, and `b_families.py` checks the full list of requested multiplicities and the weighted-ring dimension formulas by independent monomial counting. The program `b_topology.py` checks the explicit rational Mayer–Vietoris boundary matrices and their nonzero determinant for arbitrary admissible twist data. The C programs compute the cohomology of every elementary double-complex summand directly from its matrices and check the fibre-intersection sign in an oriented real period frame. The D programs verify root-line cohomology, equivariant truncated Koszul ranks and division-polynomial identities. These finite checks accompany the proofs; agreement with an index does not substitute for a sheaf calculation.

For the period scalar the working precision is one hundred decimal digits. The saved output records the modular Fourier evaluation and an independent real-period quadrature, the Newton

iteration on the Weierstrass equation, its residual and the approach parameter. The forty-digit values printed in the paper are conservatively truncated. The proof of nonvanishing uses the good-reduction elliptic curve and its marked Abel coordinate, so it does not infer nonvanishing from rounded decimal output. The order-three leading coefficient is independently checked using modular Fourier and theta evaluations.

The numerical outputs do not contain rigorous interval enclosures. In particular they certify agreement at the stated working precision, not a machine-verified error bound. Every asserted nonvanishing used in a proof has a separate exact argument. We retain the two distinct notions of a global representative defined by an elliptic Green operator and a finite atlas of coefficient functions; the latter calculation for the representative Q is explicitly unfinished in C3.8.

We searched on 12 November 2026 for the exact arXiv identifiers in the prompt and for combinations of "complex six-spheres", "Hodge", "tangent cohomology", "Kuranishi", "deformations" and "locally miniversal", including searches restricted to the recent forty days. We opened the attached papers of Engel (2026), Viaclovsky (2026b,a,c), Liu and Rollenske (2026), Yang (2026) and Wang (2026). The last attached version of Wang's paper is dated 8 November 2026. The other singular-fibre configurations in those papers are not covered by the unchanged-local-model comparison of Part B.

The search also located two materially different versions of arXiv:2610.00250. Version 1, Du et al. (2026a), contains tangent cohomology and deformation claims in a 107-page manuscript; version 2, Du et al. (2026b), is a 30-page paper whose stated scope is Oka geometry and whose deformation sections are absent. We inspected both. The root-base method in Part D was considered in light of the earlier version, but every bundle, character, cusp correction and deformation conclusion used here is proved in this dossier. We do not promote a superseded claim to a theorem by citation, and we do not infer from its omission that it is false.

The remaining comb data in E2.3 depend on an actual smoothed period section. The existence theorem does not specify one numerical set of transition matrices for X_{12}. We have computed the universal degree expressions, all four tangent dimensions, and the middle Hodge growth bounds, but have not constructed and solved those smoothing equations. Accordingly the report marks that requested item unfinished; it does not turn the explicit bundle kernel into an unevaluated numerical answer labelled complete.

References

Angella, D. (2018). Hodge numbers of a hypothetical complex structure on S^6. *Differential Geometry and its Applications*, 57:105–120. verified: attached arXiv v2 PDF opened; Section 4 and displayed relations; publication metadata checked.

Claude and Alpöge, L. (2026). The (3,4,infinity) modular family of 2-tori, completed at its three special points, is a complex structure on S^6. 108-page manuscript. verified: supplied original PDF opened; transcription checked against original pages 5, 62–63, and 81; AI-produced source is not treated as independent proof.

Du, Y.-H., Guo, B., Kusakabe, Y., and Xie, S.-Y. (2026a). Oka geometry and deformations of torus-fibred complex six-spheres. arXiv:2610.00250v1, superseded version. verified: version record and 107-page PDF opened, especially Section 6; deformation claims occur in v1 and are not in the current v2; not used as an established theorem.

Du, Y.-H., Guo, B., Kusakabe, Y., and Xie, S.-Y. (2026b). Oka geometry of complex six-spheres. arXiv:2610.00250v2. verified: version record and 30-page PDF opened; revised 23 October 2026; used only to document the scope of the current version.

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Viaclovsky, J. (2026a). Combs and complex structures on the six-sphere. arXiv:2610.10451v1. verified: attached full PDF opened; Theorem 1.1, Theorem 2.3, Propositions 3.2, 6.1, 8.3, 9.1, and Sections 9.3–10.

Viaclovsky, J. (2026b). Logarithmic transformations on the six-sphere. arXiv:2610.39102v1. verified: attached full PDF opened; Theorem 1.1, Proposition 2.3 and Section 1.1; version dated 29 October 2026.

Viaclovsky, J. (2026c). A two-parameter family of complex structures on S^6. arXiv:2609.33785v1. verified: attached PDF opened; introduction and Section 6.2; its extra rank-one degenerations are outside the unchanged-fibre comparison here.

Wang, Z. (2026). A complex structure on a rational homology six-sphere with two-torsion. arXiv:2609.11541v2. verified: attached PDF opened; abstract and main topology theorem; version dated 8 November 2026.

Yang, X. (2026). Compact complex threefolds fibred by two-dimensional complex tori. arXiv:2609.29666v1. verified: attached PDF opened; introduction, multiplicities and classification statements; these different monodromy pairs are not silently included in Part B.